
O-RAN Work Group 1 (Use Cases and Overall Architecture)

RIC Enabling NTN Deployments

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Foreword

This Technical Report (TR) has been produced by WG1 of the O-RAN ALLIANCE.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN ALLIANCE modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

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Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

Introduction

Support for Non-Terrestrial Networks (NTN) has been introduced in the 3GPP specifications starting from Release 17. Release 18 has been recently completed, with further enhancements in development as part of Release 19 and expected in the upcoming Release 20.

NTN has gained substantial momentum in the telecommunication industry, with the first services in support of IoT and direct-to-device messaging launched, and more deployments including NR-NTN upcoming. NTN presents an emerging opportunity for Open RAN and there are multiple research/development projects and evaluations in progress focusing on NTN implementation leveraging Open RAN technologies. Non-Terrestrial Networks, in particular NGSO constellations, dynamic GSO, as well as hybrid multi-orbit constellations, present complex architectural, control, radio resource management, mobility and orchestration challenges which could be addressed by Open RAN technologies.

The O-RAN ALLIANCE is currently developing specifications for O-RAN RIC-enabled optimization and autonomous management of terrestrial networks. Standardization of O-RAN RIC-enabled optimization and autonomous management for NTN and converged NTN+TN interworking deployment scenarios is essential for the NTN operator constellation planning and architectural phases, as a key enabler for their commercial success.

This intelligent management of the RAN and its operation is naturally addressed by the Non- and Near-RT RICs. However, while there are many use cases documented for their employment in terrestrial 5G networks, moving the network functions into space brings in a whole new set of challenges and constraints, and opens up completely different novel implementation approaches for the existing O-RAN use cases (such as TS and QoS optimization), as well as a new host of use cases that are specific to NTN. For example to account for the new control signaling features introduced for NTN in 3GPP Releases 17 and 18 (such as SIB19 and timer/distance-based CHO), as well as to address challenges unique to NTN.

This technical report investigates use cases applicable to NTN and presents a gap analysis on possible impacts to WG2 and WG3 specifications of the O-RAN ALLIANCE, in preparation for normative work in WG2 and WG3. Two main groups of use cases are identified:

- **Use Case Group #1:** How the TN and NTN segments will intelligently cooperate and provide a seamless experience to the users as they traverse between TN and NTN, and vice-versa, while making sure that the available spectral resources are optimally utilized.
- **Use Case Group #2:** Once in the NTN segment, how are the users and their traffic flows going to be intelligently managed by the SNOs such that they are assuring the types of services being sold while accounting for temporospatial variations in motion of the space and aerial assets, weather, spectrum availability, onboard power and hardware constraints, different terminal classes, fast mobility, Line of Sight (LoS), QoS, etc.

1 Scope

The present document specifies a technical report on RIC enabling NTN deployments use cases.

2 References

2.1 Normative references

Not applicable.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

[i.1] Non-Terrestrial Networks (NTN)

NOTE: Available at: <https://www.3gpp.org/technologies/ntn-overview>

[i.2] 3GPP TR 38.821: “Solutions for NR to support Non-Terrestrial Networks (NTN)”

3 Definition of terms, symbols and abbreviations

3.1 Terms

Void

3.2 Symbols

Void

3.3 Abbreviations

For the purposes of the present document, the abbreviations [given in [i.1] and the following] apply:

CHO	Conditional Handover
EFC	Earth Fixed Cell
EMC	Earth Moving Cell
FDD	Frequency Division Duplex
FR1	Frequency Range 1
GSO	Non-Geostationary Orbit
HO	Handover

IoT-NTN	Internet of Things Non-Terrestrial Networking
ISL	Inter-Satellite Link
KPI	Key Performance Indicator
LEO	Low Earth Orbit
LoS	Line of Sight
NB-IoT	Narrow Band Internet of Things
NR-NTN	New Radio Non-Terrestrial Networking
NTN	Non-Terrestrial Network
NGSO	Non-Geostationary Orbit
OTA	Over-The-Air
PCI	Physical Cell Identity
QEFC	Quasi-Earth-Fixed Cell
RIC	Radio Intelligent Controller
SMTTC	SSB Measurement Timing Configuration
SNO	Satellite Network Operator
SSB	Synchronization Signal Block
TDD	Time Division Duplex
TN	Terrestrial Network
UAV	Unmanned Aerial Vehicle
UE	User Equipment
UML	Unified Modeling Language

4 NTN development status in standard organizations

4.1 3GPP

4.1.1 Overview

Non-Terrestrial Networks (NTN) are a significant focus within the 3GPP standards, particularly for integrating satellite into 5G and beyond. Here's a summary of the current development status:

1. Release 15 and 16: Study phase on scenarios, use cases, channel models and impact of NTN to the 5G System, as well as solutions for 5G New Radio (NR) support for Non-Terrestrial Networks.
2. Release 17 and 18: These releases introduced NTN with a transparent (bent-pipe) architecture, where satellites act as radio repeaters without on-board processing capabilities. This approach was chosen for its simplicity and early deployment potential. Both NR-NTN and IoT-NTN are supported. Release 18 introduced support for NR NTN above 10 GHz, such as broadband in Ka band.
3. Release 19: This release marks a major milestone with the introduction of a regenerative payload architecture. In this scenario, a complete gNB (Next Generation Node B) is placed on the satellite, allowing for greater integration and flexibility, and opening up new constellation topologies and use cases). This architecture supports packet processing directly on the satellite, enabling features like improved mobility management.
4. Release 20: The next release is currently in the planning phase and a 3GPP RAN workshop in RAN#106 (12/2024) discussed potential NTN features for Rel-20. The initial work plan for Rel-20 was decided in RAN#108 (06/2025).

Technological Challenges: The development of NTN involves unique challenges such as high Doppler shifts, longer propagation delays, and larger cell sizes due to the high altitudes of satellites. Enhanced beam management and regulatory considerations are also crucial. Most of these challenges are already covered by Rel-17 and 18.

Future Prospects: The ongoing work aims to further integrate NTN with terrestrial networks, enhancing global coverage and resilience. The regenerative payload architecture is expected to support advanced functionalities like "store and forward," which allows satellites to store data temporarily and forward it when connectivity is restored, and also UE-satellite-UE communication with reduced latency.

4.1.2 R17/R18/R19 work items for NTN

Release 17:

1. Transparent payload architecture with FDD systems.
2. Adaptation to physical and access layer aspects, radio access network, system architecture, radio resource management, and RF requirements.
3. Support for NTN-NR (5G New Radio (NR)) based satellite access in FR1 bands for handheld devices.
4. Support for NTN-IoT (NB-IoT and eMTC) based satellite access for massive IoT and handheld messaging use cases.

Release 18:

1. Enhancements for NR-NTN and IoT-NTN, especially the time and location based conditional handover (CHO) for mobility enhancements.
2. Enablers for NR based satellite access in bands above 10 GHz (including introduction of Ka-band, see 3GPP TS 38.108 and 38.101-5).
3. Support for fixed and moving platforms (e.g., aircraft, vessels, UAVs) and building-mounted devices.

Release 19

1. Downlink Coverage Enhancements with CD-SSB periodicity relaxation and enhanced beam-hopping support
2. Uplink capacity enhancements.
3. Support for a non-terrestrial network architecture with 5G system functions on board the NTN vehicle (regenerative payloads), including the possibility for inter-satellite links (ISL).
4. Use of RedCap devices within FR1 NTN.
5. Store & Forward (S&F) operation based on regenerative payload to support non-complete constellations.
6. Specification of Ku-band for NTN operations
7. Support for half duplex operation in TDD and FDD scenarios for IoT-NTN
8. New L- and S- frequency band definitions
9. Over-The-Air (OTA) antenna requirements and performance framework for NTN

4.2 O-RAN

4.2.1 Overview

To the group's best knowledge, no approved contributions or normative work on NTN exist so far in O-RAN ALLIANCE, but there have been some requests for the definition of new WIs and research areas in nGRG.

4.2.2 Study and work items for NTN

The following proposals have been submitted so far:

1. New feature work for NTN deployments enabled by RIC.
2. Study on open TN-NTN architecture based on O-RAN
3. Research for requirements on O-RAN based satellite clustering for NTN in nGRG
4. Work for regenerative NTN support in O-RAN in WG1 UCTG

5 Key NTN use cases analysis in O-RAN enabled by RIC

5.1 Use Case Group 1: TN and NTN interworking and cooperation

5.1.1 Overview

In most cases, these NTN networks and services are not anticipated to be completely standalone offerings, but rather intended at least in part as a complement to Terrestrial Networks, for example to provide coverage extension in white spots and hard to reach areas, 3D coverage at altitude, or a layer of resiliency. In fact, the ability for NTN to eliminate or alleviate coverage gaps in the Terrestrial Network was one of the key motivations for the introduction of NTN in 3GPP since the beginning.

Based on the current market trends, it appears NTN deployment motivations will in most cases fall into one of the following categories:

- Existing Satellite Network Operators (SNOs) who are looking to serve the NTN market in addition to or in evolution of their existing non-3GPP satellite services.
- New entrant SNOs who are seeking to uniquely provide NTN services.
- Mobile Network Operators (MNOs) who are seeking to extend their network via NTN.
- New entrant MNOs who are seeking provide homogenous TN and NTN coverage

Therefore, it is expected that many MNOs may seek to extend their own Terrestrial Network services via NTN, however the current market trends indicate that they will likely not seek to deploy and operate their own satellites, but rather form agreements with one or multiple NTN SNOs, who will provide coverage over areas of interest on behalf of their partner MNOs.

For this to be achieved, the MNO's and SNO's networks will need to interact and coordinate with one another at some level. The required degree of interaction and information sharing required between the two operator domains will depend on the requirements of the specific use case and on the depth of integration between the two segments. This interaction is essential in order to for example make the NTN network aware of where the boundaries of the terrestrial network exist, share handover preparation information, reselection targets, etc. and conversely make the terrestrial network aware of when an NTN is available to its users.

The following clauses describe some of the use cases that fall within the category of TN and NTN interworking and cooperation and that have been grouped under Use Case Group 1 (UCG#1).

5.1.2 Use Case 1-1: TN-NTN Coverage Information Sharing

5.1.2.1 Use Case Description

In order for an MNO and their partner SNO(s) to be able to support efficient complementary coverage via NTN, the TN network may need to be aware of the available NTN coverage footprint, and vice versa. This can include current coverage (i.e. already active) as well as potential coverage (i.e. not yet active, but possible), and will need to include some details of the coverage to be provided (e.g. applicable NTN cells, their configuration and corresponding serving satellite ephemeris).

Depending on a number of operational factors and system constraints, such as feeder link availability, power and radio resource availability of the satellites forming the NTN constellation, the active cells, allocated resources (and thus capacity), as well as the specific coverage areas that an NTN operator chooses to serve or not serve may change over time. In addition, due to orbital precession, the specific satellite(s) serving a given cell coverage area on the ground at any point of time (e.g. in a scenario with Earth-fixed or quasi-Earth-fixed Cells) also change over time. Thus there is a need for the NTN segment to be able to communicate this information to the TN segment, such that the TN segment can be aware of which NTN cells are available or might become available in the future, as well as update the information being broadcast in SIB19/SIB31 messages in terrestrial edge cells.

Likewise, in the reverse direction, an NTN operator may normally not be aware of the footprint, topology, configuration or even presence of terrestrial networks within the coverage of an NTN, unless this information has been separately sourced or disclosed by the MNO. Whilst some of this information can be shared at network planning by the TN MNO to the NTN SNO,

the TN MNO network is expected to evolve over time, potentially also dynamically via the introduction of network energy saving (NES) techniques, and thus may need to be updated at runtime.

In scenarios where the NTN coverage may totally or partially overlap with the TN coverage or its edge, by having awareness of the TN footprint or of its edge, the NTN operator may be able to selectively deactivate NTN coverage in overlapping areas, in order to save system power, capacity, or to enable coverage in other locations. In scenarios with no overlap (e.g. due to TN spectrum reuse in space), the NTN operator may need to adjust the NTN coverage edge to match as closely as possible the TN coverage edge, at least for certain frequencies.

5.1.2.2 Problem statement and challenges

In order to fulfil the requirements of these scenarios, the TN and NTN segments would need to have a method of exchanging at least the minimum necessary information on their respective coverage footprints, required to fulfil complementary coverage functionalities. The current 3GPP- and O-RAN-specified interfaces between operator domains, such as the roaming interfaces and other interfaces at the Core network and RAN level (e.g. Xn), do not support sharing of this type of information.

Sharing of neighbor cell information, including neighbor relationships, and other coverage related information within a single network domain (e.g. within a TN or an NTN) is typically expected at least in part to be addressed by interfaces such as X2/Xn. However, in the interaction between TN and NTN this is either not sufficient (e.g. due to lack of the necessary primitives), or outright not possible, such as in the scenario where SNO and MNO, with both networks assumed to be aligned with 3GPP and O-RAN architecture, are two distinct operator domains, where inter-operator Xn is currently not contemplated (or implemented) and presents potential feasibility issues. In addition, both the TN footprint, topology and configuration, as well as the NTN footprint, topology and configuration, are expected to be continuously evolving and dynamic in nature, especially as operators (MNOs and SNOs alike) increasingly target maximization of energy and deployment efficiencies. Therefore, traditional manual or static mechanisms of sharing this information are not going to be fit for purpose, and there is instead a need for automated mechanisms that can dynamically provide updates on the changes to the coverage in a timely manner (order of seconds at least), as soon as they happen. It is also important that this sharing of information is performed securely and with a fine-grained level of control on the type and amount of information that is respectively shared, at any given instant of time.

Therefore, there is a gap and an identified need for a secure, programmatic and reactive mechanism for TN and NTN networks in different operator domains to interface and interact, for the purpose of selectively sharing coverage related information.

5.1.3 Use Case 1-2: TN-NTN Mobility and Service Continuity

5.1.3.1 Use Case Description

This use case covers the support for mobility and service continuity between a terrestrial network (TN) and a Non-Terrestrial Network (NTN). One of the key value propositions for Non-Terrestrial Networks is to enable complementary coverage for terrestrial networks, and to help MNOs fulfil coverage gaps. As such, enabling or facilitating mobility and service continuity between the TN and the NTN and vice-versa is a key requirement.

Whilst 3GPP defines the procedures at RAN level for TN-NTN idle and connected mode mobility and stipulates the need for sharing of coverage and satellite assistance information (e.g. Ephemeris), the 3GPP specifications do not explicitly address how this information is shared, and further how these procedures are coordinated when the TN and NTN are in different operator domains. This use case thus describes the requirements, expected procedure and possible solution to address these remaining aspects in respect to TN-NTN mobility and service continuity.

In order to be able to support TN-NTN mobility, the TN RAN needs to be aware of the NTN coverage footprint, and specifically which satellite(s) are or will be serving a given coverage area at any point in time, and which are or may be the applicable neighbor NTN cells, with their respective configurations. In 3GPP NTN Rel-17+ systems, some of this information, alongside other details such as Ephemeris of the satellites, is expected as per the 3GPP specifications to be delivered via SIB19 or SIB31 into the TN cells bordering or overlapping with the NTN coverage area. However, in order for the information to be delivered to the TN cells, the terrestrial network has to first be aware of the information in the first place. This information may in many cases need to come from an external source to the MNO network (e.g. an SNO network). In addition, there may be another scenario in which an MNO may wish to provide

Additionally, when it comes to evaluating the candidate handover or reselection opportunities to an NTN cell from a TN cell, there may be a subset of satellites which are better suited as target candidates, and there is a need to study how the O-RAN RIC/SMO architecture might be leveraged in optimizing the selection, for example by tailoring the SMTC list and the NTN-

NeighCellConfigList provided in SIB19 to include only the satellite candidates that actually make for good handover or reselection targets, in some cases based on the specific UE service requirements on RAN and line-of-sight (LoS) conditions.

5.1.3.2 Problem statement and challenges

Whilst 3GPP specifies mechanisms for TN-NTN mobility and service continuity, many of the capabilities and information exchanges required to implement said mobility, and to more in general facilitate the seamless interworking between TN and NTN are currently not addressed by 3GPP specifications, and could on the other hand be effectively addressed by O-RAN.

In order to support mobility from TN to NTN, the TN network must be aware at least of the following information, and their applicability in time:

- Satellite assistance information (e.g. Ephemeris) of the target NTN systems, satellites and cells available for handover or reselection
- Cell topology and availability of the NTN cells neighboring or overlapping the TN coverage edge, that could be candidate target cells for measurement, reselection or handover
- Cell configuration of the NTN cells neighboring or overlapping the TN coverage edge, necessary for measurement configurations, reselection policy configuration and handover triggering

In addition, the network-initiated Connected Mode handovers must also be explicitly triggered by the network, which must be able to:

- Request measurements to the target NTN network, to identify HO cell candidates
- Request HO triggering to the target gNB

In a terrestrial network scenario or an intra-operator TN-NTN scenario, many of these capabilities are or can be enabled by the X2/Xn interface. However, X2/Xn may not be feasible in an inter-domain/inter-operator environment.

In addition, given that between two operators, the network topology might be partially or completely obfuscated, there is no way for the Source gNB to even identify which target gNB endpoint (e.g. gNB-CU) to address the measurement and handover requests, as the X2/Xn interface exists between individual eNBs/gNBs.

Thus, there would be no mechanisms for signaling of neighbour relationships, including for measurement configurations. Furthermore, the TN may not even be aware of the neighbour NTN cells (esp. for candidate cells that are not active, in systems deploying selective coverage and/or beam hopping).

Therefore, these challenges arise with this scenario and the lack of a suitable interface, for example:

- No awareness by the TN of the target NTN cell configuration (e.g. frequencies, PCI, etc) required to configure measurements, HO triggers, Reselection policies.
- No awareness of the gNB topology in the target network (either TN or NTN), and thus no awareness of which gNB to even establish an Xn connection or trigger an HO request or measurement request.
- With 3GPP NTN R17+, the TN might not be aware of Satellite Ephemeris (SIB19/SIB31 content), required by the UE to perform UL pre-compensation
- The NTN coverage may not be active (or may be configured with insufficient duty cycle/capacity) in the target area that the UEs are about to move to (satellite coverage is limited and must be selective)
- The NTN may not have any awareness of incoming load, QoS requirements or UE capabilities
- In 3GPP R19 NTN, SSB Periodicity could be relaxed (e.g. up to 160ms). This can impact UE sync performance/reliability and may need specific SMTC configuration for measurement before TN to NTN handover
- UE capability signaling is very heavy. Forcing the NTN to perform UE capability exchange from scratch may have adverse impact on NTN capacity
- Letting the UE/network declare RLF and Detach the UE, would lead to a network selection/access from scratch, which creates service interruption and unnecessary signalling load.
- The NTN network may not be aware of which (or any) UE in TN are about to be handed over (or reselect to) NTN

- In the reverse direction (NTN to TN), no awareness by the NTN of the target TN cell configuration (e.g. frequencies, PCI, etc) required to configure measurements, HO triggers, Reselection policies

5.1.4 Use Case 1-3: Request of NTN coverage for Disaster Response

5.1.4.1 Use Case Description

In some cases a Terrestrial MNO might wish to explicitly request NTN coverage in a given area, either to fulfil a coverage gap, or to respond to a total or partial outage of the terrestrial network, such as in a disaster scenario. This same capability could also be used simply to request the NTN to allocate additional capacity in a given area, in order to respond to an expected or witnessed increase in demand.

Scenario 1) TN MNO1 identifies a coverage gap in its coverage, or is interested in providing additional coverage overlay, and thus requests NTN coverage to specific target geographical area to NTN SNO1.

Scenario 2) TN MNO1 already has active NTN coverage by NTN SNO1 in a given target geographical area, but identifies a need to allocate additional capacity (e.g. due to an upcoming event).

Scenario 3) During a disaster scenario (e.g. a hurricane), cell towers of TN MNO1 in a given area are affected by an outage (e.g. power outage) and are thus unavailable. TN MNO1 requests NTN coverage (in dedicated or shared spectrum) to the affected target geographical area to NTN SNO1.

The scenarios from 1 to 3 are applicable also when an MNO has a commercial relationship or wishes to establish a commercial relationship with more than one NTN SNO partner.

In addition to the basic coverage or capacity request, a TN MNO may wish to communicate additional information to their NTN SNO partner, for example:

- Expected load or last-known load
- Required capacity
- QoS information and/or Slice policies for specific services that may be required (e.g. public safety, emergency messaging and/or calling)
- List of UEs for prioritization/QoS policy
- Pre-sharing of UE capabilities mapped to explicit list of UEs
- Interference thresholds and coverage edge contours or other constraints (e.g. to prevent interference, in case of terrestrial spectrum reuse)
- Desired cell configuration parameters
- In case of terrestrial spectrum reuse or other form of spectrum lease, information on the spectrum available for deployment and any configuration requirements.

5.1.4.1 Problem Statement

Current methods for an MNO to request NTN coverage from an SNO are entirely reliant on traditional office IT communication (e.g. emails, meetings), and require extensive discussions and pre-agreements not only at the commercial level, but also at the technical level, as a pre-requisite.

With current methods, derivation of technical parameters for requests of either complementary coverage or capacity rely on extensive offline simulations performed by both parties respectively. These simulations are usually based on abstracted knowledge of the network, and relatively limited advance knowledge of the respective systems, also due to confidentiality constraints. From these simulations and plans, a notional NTN network plan is usually derived and agreed by both parties, via extensive discussion. In some cases, the simulations used to derive the plan may not even be fully reflective of the up-to-date conditions of the network (mostly because they are generated not from the live network, but from offline system-level simulator toolsets), nor of the real-world practical conditions in which the system will operate.

In addition, any subsequent update usually needs to go through a similar formal cycle of discussions, simulations, agreements and formal approvals, de-facto preventing the enactment of any reactivity to changing requirements or events in responsive timeframes. In some cases, due to the relatively “static” nature of the plan and of the need to pre-plan in advance, often with a high level of detail, this can also lead to inflexibility in any regulatory filings submitted, where exact static values may be specified or filed for, and thus require an update to the regulatory filings in case some parameters are altered or optimized.

As a consequence, the resulting plan exhibits a few unfortunate byproducts:

- 1) **Staleness:** The NTN network plan derived is relatively static and represents a snapshot of the respective systems at a given moment of time, which may become stale over time, due to being unable to automatically reflect any subsequent changes or optimizations in either the MNO terrestrial network coverage, or the SNO's NTN coverage.
- 2) **Immutability and lack of reactivity:** Changes in the NTN network plan are difficult and slow to introduce, and not only it cannot easily accommodate either changes in the actual respective network topologies, capabilities and plans, but it is also unable to be adapted in responsive time frames, to react to changing or unforeseen needs by either SNO or MNO, such as need to provide Disaster Coverage in response to an outage, spikes in demand or other time-sensitive events.
- 3) **Poor accuracy:** Due to the simulation environments used to derive the technical parameters of the plan, and due to issues with confidentiality, the plan may not reflect the accurate and up-to-date state, capabilities or parameters of the live network, reducing the accuracy of representation. Its accuracy also to the physical live environment and its interaction with terrain, weather and other factors is also very limited.
- 4) **Conservativeness and sub-optimality:** Due to the above issues, the plans are usually defined with relatively (sometimes extremely) conservative margins, in order to avoid, or at least reduce, any risks of disruption to user experience. Inevitably, this also results in significant sub-optimality.

These issues could be resolved via a form of automated and programmatic mechanism, enabling the exchange of requests (as well as advertisements of availability) for coverage and capacity between different entities, such as for example between one MNO and one or more SNOs to provide NTN coverage and capacity in reaction to changing demand or to an unforeseen even, such as a disaster scenario.

There are currently no existing interfaces, APIs or otherwise programmatic and/or automated mechanisms to support either static, semi-static or let alone dynamic requests for coverage and/or capacity between different operator entities within either the 3GPP or the O-RAN specifications. Such requirements and concepts have been raised and discussed from time to time, at least within 3GPP in the context of NTN, but have never so far been introduced. In particular, there is currently no satisfactory mechanism to satisfy certain mission-critical use cases for the public good, such as respond to the need to supply Disaster Coverage via NTN in response to total or partial outages to parts or totality of an MNO's terrestrial network within meaningfully responsive timeframes from the occurrence of an event (e.g. seconds or at most minutes).

5.1.5 Use Case Group 1: Summary, Proposed Solution, Benefits and Impact to the O-RAN Architecture

5.1.5.1 Summary of Gap Analysis

The previous clauses describe a number of use cases that involve interaction between the TN and NTN networks and summarize the current gaps and required interactions. Due to the commonalities between the different use cases, it was decided to address possible solutions in a single, common clause. The following clause thus analyzes possible common solutions to address the aforementioned use cases. From analyzing the different use cases in Group 1, we can also summarize a list of requirements that a possible solution must be able to fulfil.

In all the previously discussed use cases of Group 1, it is assumed that the TN and NTN segments could be either part of the same operator domain or, more typically, part of two different and independent operator domains. It is also possible that, within a single operator domain, the operator may choose to maintain domain separation between the TN and NTN segments. Any new solution(s) must be capable of addressing all such cases.

In addition, the solution should not be limited to a one-to-one relationship between TN and NTN segments, but should also enable one-to-many relationships, for example to support the case where an MNO may have multiple SNO partners, and vice-versa the case where an SNO may have multiple MNO partners/customers, but also the case where multiple separate TN and NTN segments exist within a single operator, as separate administrative domains.

The key requirements for the solution can thus be summarized as follows:

- 1) Allow interaction between TN and NTN segments in the same operator, assigned to a single administrative domain
- 2) Allow interaction between TN and NTN segments in the same operator, assigned to multiple separate administrative domains
- 3) Allow interaction between TN and NTN segments in different operators and independent operator administrative domains

- 4) Allow interaction between one TN segment and multiple NTN segments (in the same or different operator domains)
- 5) Allow interaction between one NTN segment and multiple TN segments (in the same or different operator domains)
- 6) Allow sharing of NTN radio cell coverage information from the NTN to the TN
- 7) Allow sharing of TN radio cell coverage information from the TN to the NTN
- 8) Allow sharing of cell configuration between the TN and NTN
- 9) Allow sharing of NTN satellite assistance information with the TN (e.g. in the form of satellite Ephemeris for SIB19/SIB31)
- 10) Allow secure sharing of information between different operator trust domains
- 11) Allow dynamically-selective sharing of information across TN and NTN based on respective operator's control
- 12) Allow advertisement of network capabilities
- 13) Allow requesting of NTN coverage to one or more target locations in 3D space and time
- 14) Allow requesting of NTN capacity to one or more existing target NTN coverage areas
- 15) Allow sharing of load information between networks
- 16) Allow sharing of UE capability information between networks
- 17) Allow sharing of QoS and Slice policy information between networks, including for specific UE to be prioritized (e.g. list of UEs)
- 18) Allow sharing of information necessary to configuration of measurements for mobility between TN and NTN (in both directions)
- 19) Allow triggering of measurements from the partner network (TN or NTN), e.g. in preparation for a mobility event
- 20) Allow triggering of mobility events for Idle and Connected mode mobility between TN and NTN (in both directions)
- 21) Allow sharing of acknowledgements and status information for previously advertised or requested capabilities

The requirements described above cannot be presently satisfied by any existing interface, architectural element or procedure in the current 3GPP or O-RAN Architecture, at least for the case where the TN and NTN segment are operated by different entities (e.g. different operators) or part of different administrative domains.

The O-RAN architecture and RIC/SMO complex, however, do offer a unique framework for management of radio coverage in an operator domain, to complement the 3GPP specification and address challenges not addressed by 3GPP, and are either already capable, or are expected to be capable (e.g. in the case of NTN) of accessing the required information in the respective operator networks. They therefore offer a unique opportunity to introduce a viable solution for TN-NTN radio network coverage level inter-working and cooperation, if a suitable mechanism to enable communication between the TN and NTN domains can be identified.

5.1.5.2 Proposed Solution: New TN-NTN Interface

In order to satisfy the requirements outlined above, for the scenario in which the TN and NTN segments are operated by different operators, an MNO and an SNO respectively, or under otherwise different administrative domains, a potential proposed solution is the introduction of a new and novel O-RAN interface or API between the two operator domains that is suitably designed to cross the inter-operator domain trust boundary.

If we assume that both the TN and NTN are respectively adopting the O-RAN architecture and thus both have their own respective Near/Non-RT RIC and SMO deployments, the new interface could be designed to exist between the MNO's and SNO's respective Near/Non-RT RICs or SMOs, facilitating different levels of information sharing. Such capability and interface are not currently specified or contemplated by the O-RAN Architecture and Specifications, and therefore it is identified as a gap. Overall, this cross-operator mode of operation and interaction was not envisaged by 3GPP or the O-RAN Alliance previously and would require investigation into how the O-RAN architecture may need to evolve, potentially with the specification of one or more new interfaces.

After several discussions on at which layer should this new interface exist, and considerations for various candidate options, it seems that the most likely suitable approach may be to specify this interface between the Non-RT RICs or SMOs. However, the possibility of supporting a new interface between the Near-RT RICs could also be considered and should be further studied if the requirements for use cases demand it.

5.1.5.3 Impact to the O-RAN Architecture

This clause of the Technical Report outlines different architecture options for TN-NTN interworking depending on whether the two segments are operated by the same or a different operator (or different administrative entities within the same operator), and considers as a solution the specification of at least a new interface between an NTN Non-RT RIC or SMO and a TN Non-RT RIC or SMO, at least for the case where TN and NTN segment are operated by different entities.

The impact can thus be summarized as follows:

- **Case 1 (TN and NTN operated by different entities):** Introduction of a new East/West API interface or service between Non-RT RICs or SMOs of the two operators.
- **Case 2 (TN and NTN operated by the same entity):** Reuse of the existing A1 interface reference point and augmentation of the A1 and the RIC/SMO layer with procedures and information relating to TN-NTN information sharing.

For the scenario in which the TN and NTN segment are operated by different operators, an MNO and an SNO respectively, or part of different administrative domains, a potential proposed solution is to introduce a new East/West API interface or service between the SNO's NTN Non-RT RIC and the MNO's TN Non-RT RIC. Via such interface, the TN Non-RT RIC and the NTN Non-RT RIC could exchange information regarding the respective coverage areas as well as other information as required for mobility and service continuity procedures.

Figure 5.1.5.3-1, below, depicts the aforementioned topology in an inter-operator scenario, which sees separate TN and NTN network segments operated by different operators (e.g. an MNO and an SNO respectively), and highlights the potential new interface reference point between Non-RT RICs of the two operators. It is important to note that today some control and management functions may still be supported outside of the O-RAN architecture framework as they are not currently covered by any O-RAN function and are opaque to the O-RAN architecture. These have been represented in the figure below as “Non-O-RAN NTN Functions” and are currently outside of the scope of O-RAN and of this Technical Report. Similarly then, the interface between the Non-O-RAN NTN Functions and NTN gNB is currently a bespoke interface, which is also outside of the scope of O-RAN and of this Technical Report.

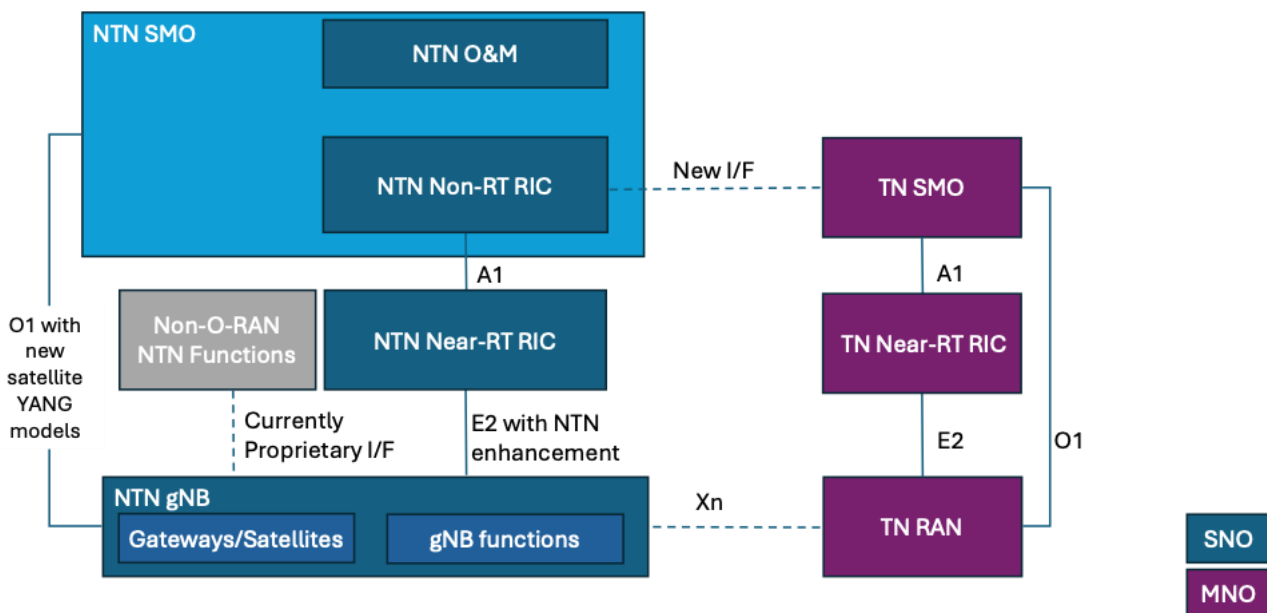


Figure 5.1.5.3-1: Topology of inter-operator scenario

For the case in which the TN and NTN segment are operated by the same operating entity, the existing architecture and interfaces could be reused, whereby a single Non-RT RIC and SMO may exist and assume the responsibility of managing both the TN and NTN segments of the network.

The existing mechanisms of interaction between the Non-RT RIC and the Near-RT RIC via the A1 interface could in principle thus be reused, with the assumption that the operator's Non-RT RIC would have multiple A1 interface associations with one or more TN Near-RT RICs, as well as one or more NTN Near-RT RICs. The common Non-RT RIC could then supply information to both the TN and NTN Near-RT RICs in the shape of A1 policies or enrichment information, with appropriate extensions to the standardized data types, yet to be determined.

Figure 5.1.5.3-2, below, depicts the aforementioned topology where both TN and NTN segments are operated by the same operator (e.g. either an MNO with an NTN component, or an SNO with a TN component). The topology of both TN and NTN segments follows the existing O-RAN architecture and reuses the existing O-RAN interfaces, such as the A1 reference point. Similarly to the inter-operator case, it is expected that today some control and management functions may still be supported outside of the O-RAN architecture framework as they are not currently covered by any O-RAN function and are opaque to the O-RAN architecture. These have been represented in the figure below as “Non-O-RAN NTN Functions” and are currently outside of the scope of O-RAN and of this Technical Report. Similarly then, the interface between the Non-O-RAN NTN Functions and NTN gNB is currently a bespoke interface, which is also outside of the scope of O-RAN and of this Technical Report.

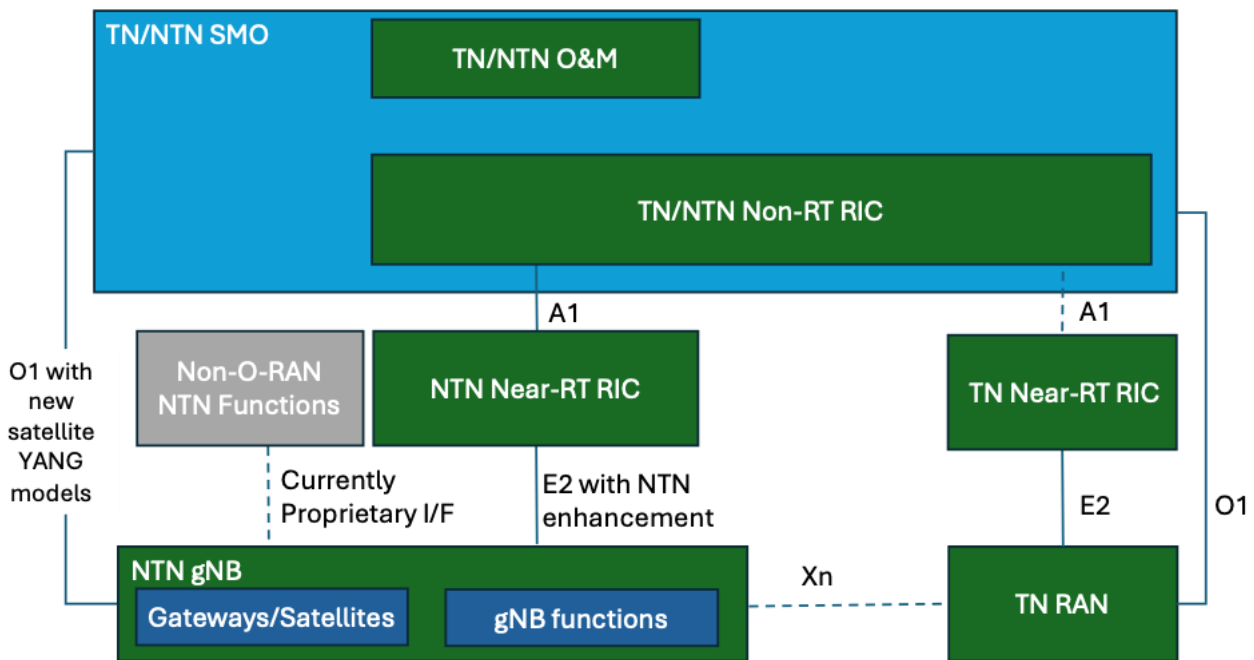


Figure 5.1.5.3-2: Topology of single operator scenario

5.1.5.4 Benefits

There are several benefits to this new proposed interaction, some of which have been captured in the Use Cases of the Use Case Group 1, and include the enablement of TN to NTN and NTN to TN mobility and service continuity scenarios, such as reselection and handover between the TN and the NTN and vice-versa, the enablement of the ability to request NTN coverage in a number of operator scenarios, as well as opportunities for higher levels of optimization, for example in how coverage is provided between the TN and NTN segment, and in ensuring truly complementary and mutually-beneficial and optimal coverage between the TN and NTN segments.

5.2 Use Case Group 2: Intra-NTN Mobility Use Cases

5.2.1 Overview

This clause describes use cases applicable within the NTN, in particular, use cases related to aspects of mobility and intra-system service continuity. Mobility events in an NTN constellation may be driven by a number of factors and impairments that are specific to NTNs, in addition to typical UE mobility requirements, such as satellite (i.e. network node) mobility as well as delay and Doppler, spectrum and power constraints and others such as line-of-sight. The following are some examples of key mobility scenarios that can be expected in an NTN system:

- Feeder link management (e.g. switching, load balancing, etc)
- Satellite mobility
- Satellite load balancing and selection
- UE Mobility
- Inter-satellite Link (ISL) mobility

3GPP Releases 17, 18 and 19 introduced new procedures specifically to address NR-NTN and IoT NTN mobility management, that are designed to take into account such NTN-specific impairments and influencing factors. However, whilst the procedures are specified in detail in the 3GPP specification, their actual implementation and the full implications from a network architecture and implementation are not stipulated and are instead left up to operator and/or vendor implementation choice.

This chapter analyses a number of mobility use cases that fall in the categories listed above and that leverage existing 3GPP procedures as of Release 18, within the context of the O-RAN architecture as a possible implementation, in a bid to efficiently allocate the limited resources of the NTN system whilst maintaining and maximizing fulfilment of user service requirements.

Lastly, an initial assessment of the possible impact of the introduction of such use cases to the O-RAN architecture and specifications is made.

5.2.2 Use Case 2-1: Connected mode NTN inter-gNB handover (HO) due to Satellite mobility

5.2.2.1 Use Case Description

This use case covers the use of an NTN near-RT RIC and xApps to optimize the implementation of inter-satellite inter-gNB Connected Mode Handover (HO) within an NTN Non-Geostationary Orbit (NGSO) constellation, exemplified by a Low Earth Orbit (LEO) constellation, driven by satellite mobility. This use case is in principle formulated in a way that is largely agnostic of the NTN gNB architecture (e.g. Transparent vs Regenerative). In this scenario, users are assumed to be practically stationary or slow moving (e.g. handheld), and the mobility events are driven by satellite motion. This scenario use case description assumes Earth-fixed cells (EFC) or quasi-Earth-fixed cells (QEFC), but is applicable also to Earth-moving cells (EMC).

Many upcoming 5G NTN constellations will be deployed in Low Earth Orbit, where a significant number of satellites are needed in order to achieve ubiquitous global coverage. LEO constellations are highly-mobile and their network configuration is continuously changing, and as such UEs may experience frequent handovers from one satellite to the next.

In addition, in dense constellations, multiple candidate satellites may be able to serve a given coverage area, introducing the need for the network to select the appropriate target satellite to serve the incoming users. This could be influenced by a number of factors, such as feeder link availability, expected satellite visibility period, maximum number of concurrent beams, on-board power limitations, UE TX power and antenna gain, line of sight, spectrum constraints and more.

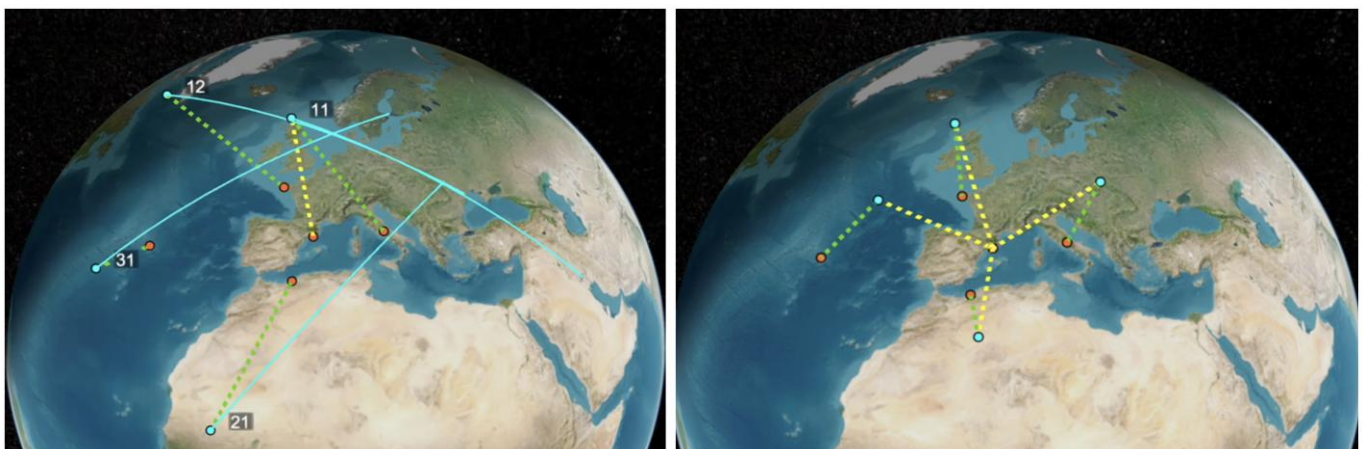


Figure 5.2.2.1-1: Example network scenario with 4 satellites from a Walker Delta formation (left) and snapshot of the moment where HO opportunity is available with 4 satellites in view (right)

The example scenario in Figure 5.2.2.1-1 shows part of an NTN LEO Walker Delta constellation with four satellites in view, two of which co-planar and two cross-planar. The projected satellite orbital trajectories are shown in cyan. As a simplifying

assumption, each satellite is associated with a single feeder link (shown in dotted green) to a given gateway (represented by red dots associated to the green dotted lines) and a single gNB broadcasting a single PCI. Feeder link mobility is thus not considered for the sake of simplicity. This allows this use case to focus on user link mobility, and makes the scenario applicable to both Transparent and Regenerative architectures.

Co-planar satellites are numbered with the same most-significant digit, whereas cross-planar satellites are assigned a different most-significant digit. The current scenario assumes that the associated satellites project a beam which is fixed to the satellite and is sweeping the Earth beneath. This eliminates the need for a system to be present which will intelligently control a steerable beam antenna that would direct energy towards a particular Earth-fixed direction, as well as maintain things like beam hopping schedules. However, in most future 5G NTN constellations, steerable beam antennas would be a key component, and their management and scheduling would have to be coordinated with the management of the 5G NTN RAN in order to achieve truly optimized operations.

At the beginning of the scenario, a UE located in Spain is currently served by satellite 11, which forms the user link to the UE (in yellow) and is supported by a feeder link (in green) to a ground station in central Italy. As shown by the orbital trajectory, in some time, satellite 11 will be approaching the end of its viable service time for the target UE (e.g. due to low elevation, scan angle or link budget). Simultaneously, the remaining three satellites 12, 21 and 31 will approach the UE coverage area and become eligible to place a beam (and a cell) in the UE's location, and they therefore offer potentially viable candidate handover opportunities. Figure 5.2.2.1-3 shows the possible candidate satellites and UE service links at the instant in which all satellites are in principle eligible to serve the coverage area.

An NTN Near-RT RIC and an associated NTN xApp could be tasked with taking all the various factors into account and driving the decision of which of the 3 available candidate handover satellites should start serving the UE once its currently serving satellite becomes unsuitable.

In addition, the previously mentioned power and spectrum constraints present a further optimization opportunity to the network and NTN xApp: for example one of the available satellites could deliver the requested level of service to the UE while consuming fewer resources than another satellite. Lastly, in a scenario with several UEs in a given coverage area, the NTN Near-RT RIC and NTN xApp must make a handover decision that accounts for the requirements and conditions of all the users in the coverage area, which may create the need for trade-off decisions.

5.2.2.2 Problem statement and challenges

The first part of the problem statement is relatively trivial. In order to support NR NTN scenarios aligned with 3GPP Release 17 and above, the NTN xApp must be able to configure the required NTN-specific measurement configurations, as well as the necessary handover triggers.

In Release 17, a new SSB Measurement Timing Configuration (SMTC) specific to NR NTN, namely SMTC4 (*SSB-MTC4List-r17*), has been introduced. In addition, the NTN cells must be able to broadcast Satellite Assistance information, in particular Satellite Ephemeris, in the form of SIB19, containing serving and neighbor satellite and cell information. Furthermore, in order to trigger a HO, the gNB must be able to include specific SIB19 content in the HO triggers.

This implicitly requires the NTN Near-RT RIC and xApp(s) to be aware of the constellation Ephemeris, the serving and target satellite timing information, as well have the ability to deliver the necessary configurations to the O-CU/DU.

The second aspect of the problem is related to coordination of the HO across multiple satellites (and cells, potentially gNBs), in order to hand the UE over to the best satellite.

Several criteria and KPIs may be taken into account to select the best target satellite, and, depending on the service requirements set by the SNO, the NTN RIC may have to either privilege one or more specific KPIs at the expense of others, or to perform a multi-constraint evaluation by ensuring adequate and balanced satisfaction of several different KPIs at once.

In terrestrial networks, assessment of handover opportunities is almost entirely focused on KPIs affecting the user link, and would typically be driven by UEs measuring the SSBs of relevant neighbour cells, and then the gNB or Near-RT RIC monitoring for SMTC-configured measurement events and reports where the quality of a neighbour cell signal would exceed that of the serving cell signal by some threshold, such as is the case in the A3 event. Measurement reporting, based on SSBs but also CSI-RS, is also very important for channel estimation in terrestrial networks to aid link adaptation, primarily due to urban deployments where the various channel state and effects of multi-path propagation are highly and frequently variable and difficult to predict, except perhaps at the cell edge. In NTN deployments, on the other hand, limited multipath effects due to focus on outdoor environments, and the predictable movement of the satellite in the sky coupled with accurate weather forecast data make it such that the user link channel state by itself is much more steady and predictable, driven mostly by satellite motion, and can thus be

quite reliably modelled and estimated. The overall link and HO candidate selection is, however, on the other hand, affected by a number of other factors, many of which are completely opaque to the gNB and cannot be addressed with measurements.

Some examples include:

- Ability to sustain the link vector for the duration of the satellite pass (or UE call flow) - influenced by a number of factors including feeder link availability
- Available on-board battery power
- Overall network load balancing goals
- Cleanest channel, and thus best achievable MCS
- Feeder link availability - e.g. due to weather or other system constraints such as antenna field of regard and LoS to the satellite.
- Line of sight (LoS) on the user link (but not only)

Feeder link availability is a particularly interesting example as it is chiefly important in transparent NTN deployments where, even in cases where the user link is in FR1, the NR-Uu waveform has to traverse also the feeder link between the satellite and the ground station, which often utilises higher frequencies (e.g Ku-, Ka-, Q- or V-band) that are notably susceptible to attenuation due to weather effects, and which makes it such that the feeder link directly impacts the NR-Uu user link radio interface. The feeder link behavior and impacts are not specified in 3GPP; this effectively makes it opaque to the gNB and its internal processes. In addition, due to the dynamic nature of the system and of the external effects (e.g. weather) that might intervene, the impact to the NR-Uu user link introduced by the feeder link might not be completely known in advance by any of the individual gNBs at the moment of handover, due to the fact that a feeder link might not even have been established yet.

Continuous satellite movement also makes it such that the channel quality is, albeit predictably, always continually and steadily changing, driven effectively by satellite motion, unlike in terrestrial networks. Not only might the different satellites and their respective feeder and user link vectors, propagating through variable weather and atmospheric conditions, result in varying channel qualities and therefore attainable MCS regimes, but they can also result in varying viable service times before either the user link or the service link becomes unsustainable, due to factors such as Line-of-Sight (LoS) or minimum elevation angles due to field-of-view (FoV) and obstructions.

Thus, serving or neighbour cell signal measurements prior to the handover can by themselves be at least partially misleading in NTN, since they do not indicate much about either the end-to-end channel quality including the feeder link and the satellite itself (e.g. on transparent systems), past and future channel state evolution throughout the satellite pass, or future vectors between each satellite and the UEs being served, or expected to be served in the future (e.g. impacted by LoS effects, such as mountain ranges which might block LoS from a given area and UEs or feeder links to a given satellite at any future point of the service pass). For example, for Satellite 31 in Figure 5.2.2.1-1, a feeder link is formed with a ground station in the Azores. As the satellite propagates due North-East, it may witness its current ground station fall out of the feeder link view before the UE falls out of the user link view, thus requiring a feeder link handover. While feeder link is not a 3GPP-specified element of the network, mobility procedures associated with it may still require intervention from the RAN, and, depending on network implementation, may potentially result in an L3 Handover as well for the UE, leading to potential service discontinuity and signaling overhead.

Furthermore, since the NTN link is often more constrained compared to a terrestrial link, an excessive reliance on user link signalling and UE measurement reports may create significant overhead on the link, negatively affecting user throughput and system capacity.

This introduces the need for intervention, and opportunity for optimization, from an external control plane that has awareness of the satellite constellation, its sub-components and its evolution, and is thus capable of predicting the behaviour of both the user links and their channel state across multiple satellites throughout their motion, as well as the impact of components of the system that are outside of the scope of any individual gNB, such, in this example, other gNBs, other satellites or NTN service nodes, or any of the gateway feeder links. In addition, to account for events in the future impacting links that have not come into existence yet, there might be a need for the NTN control plane to be able to be spatially-aware, in addition to it being temporally-aware, to propagate predict where the NTN service nodes (e.g. satellites, but not only) might be at any future point in time and which links they might be able to form.

5.2.2.3 Proposed Solution

A possible solution could be leveraging the NTN RIC, either as a combination of the Near-RT and the Non-RT RIC, or via the Near-RT RIC alone, with the introduction of a new control loop provided by one or more NTN xApp(s). Such xApps would leverage the holistic view of the satellite network provided by the NTN RIC, combined with predictive capabilities, to produce handover decisions which take into account multiple concurring factors, as well as reducing the reliance on signal measurements and reporting.

The NTN RIC and xApp(s)/rApps are configured with a digital twin of the NTN network, including, not only the 3GPP RAN segment, but also the NTN ground and space segments (e.g. the gateways, the feeder links, the satellites, etc) via which the NTN service is delivered, even though some of these elements are presently not within the scope or control of the existing O-RAN reference framework.

It is assumed that the NTN RIC may be capable of predicting NTN network node state, motion, trajectory (e.g. orbit) and behaviour in time and space. It is also assumed that the NTN RIC may be able to ingest external data sources, such as, but not limited to, weather data, to predict impact of external events on the active or candidate links, in order to aid the control loop affecting handover decisions.

By knowing the predictable orbital movement of the candidate satellites, their association with ground stations and feeder links, the relatively static position (compared to the satellites) of UE or groups of UE in a given target coverage area (e.g. cell or cell portion) and the impact of weather effects, an NTN xApp could drive the decision of which satellite to hand over the UE to by modelling the candidate link vectors which could be formed and sustained until the need for the next handover, and aiming to choose the one that satisfies the appropriate combination of constraints. The use of prediction can be considered a supplement to, not a replacement of the evaluation of UE measurement reports.

Table 5.2.2.3-1: NTN handover management

Use Case Stage	Detailed Information
Goal	Find best handover candidate satellites/cells for a UE to address impact of satellite motion and constellation evolution, while considering a number of diverse constraints and dependencies not directly visible through measurements, and trigger a Connected Mode Handover (HO).
Actors and Roles	NTN xApp, Near-RT RIC, E2 Nodes, UE.
Assumptions	All relevant functions and components are instantiated. The NTN RIC is aware of the state and motion of the constellation and is capable of predicting its evolution in time and space.
Pre-conditions	E2 interface connectivity is established, UE is in active RRC Connected Mode state to a given serving satellite (e.g. Satellite 11)
Step 1 (M)	NTN Handover Optimization xApp/Near-RT RIC subscribes to UE information change and message copy of UE mobility from E2 Nodes to understand the state of the network and the UEs within it.
Step 2 (M)	NTN Handover Optimization xApp/Near-RT RIC requests to the E2 Nodes for the UEs to perform measurements of both serving and visible candidate target cells across satellites, and configure in the E2 Nodes UE information change and message copy of UE mobility to be reported back to the NTN Handover Optimization xApp /Near-RT RIC.
Step 3 (M)	E2 Nodes configure target measurements to UEs and UEs perform measurements and monitor for configured measurement events, upon which they signal a measurement report to the serving NTN E2 Node
Step 4 (M)	NTN E2 Nodes forward the measurement report to NTN Handover Optimization xApp/Near-RT RIC
Step 5 (M)	xApp with digital twin analyses the currently employed Ground Station and transparent LEO satellite to derive the remaining service time before a handover must occur. It also evaluates the candidate handover opportunities via available neighbouring satellites and feeder links to model the link performances attainable during the service times that they could respectively offer to the UE, and compares them with the results of the measurement reports. Its output is a selection of the neighbour satellite/cell which is best positioned to serve the UEs in the target coverage area.
Step 6 (M)	NTN Handover Optimization xApp/Near-RT RIC signals to NTN E2 Node to establish the correct handover control to the target UEs, to the selected target satellites, including the correct SIB19 contents (e.g. Ephemeris) for the respective target satellites to be included in the HO request message by the E2 Node.

Step 7 (M)

NTN E2 Node implements RRC Connected Mode handover as specified by NTN Handover Optimization xApp/Near-RT RIC

```
@startuml handover optimization
title NTN Handover Optimization Use Case
Skin rose
skinparam sequenceMessageAlign left
skinparam ParticipantPadding 5
skinparam BoxPadding 10
skinparam defaultFontSize 12

box "NTN Segment" #cyan
participant "NTN HO Optimization xApp/Near-RT RIC" as xappnearrrtric
participant "NTN E2 Nodes" as e2nodes
end box

autonumber

group #White NTN Handover Optimization xApp/Near-RT RIC collect RAN metrics
xappnearrrtric -> e2nodes: <E2> Subscribe to UE information change and message copy of measurement reports
e2nodes -> xappnearrrtric: <E2> Report UE information change and message copy of measurements events
end group

group #White NTN Handover Optimization xApp/Near-RT RIC establish signal measurement and reporting config for candidate handover opportunities
xappnearrrtric -> e2nodes: <E2> Set up signal measurement and reporting configuration per-UE/per-cell
note over e2nodes: Configure measurement control for candidate serving satellites/cells and\nUE report measurement result of candidate serving satellites/cells to E2 nodes
e2nodes -> xappnearrrtric: <E2> Report the measurement result
end group

note over xappnearrrtric
xApp with digital twin analyses the currently employed
Ground Station and transparent LEO satellite to derive
the remaining service time before a handover must occur.
It also evaluates the candidate handover opportunities
via available neighbouring satellites and feeder links
to model the link performances attainable during the service
times that they could respectively offer to the UE, and
compares them with the results of the measurement reports.
Its output is a selection of the neighbour satellite/cell
which is best positioned to serve the UEs in the
target coverage area.
end note

note over xappnearrrtric : xApp derives decision to handover\nUE to Neighboring X LEO, served\nby Neighboring X gNB/NTN GW

xappnearrrtric -> e2nodes: <E2> Per-UE network-initiated HO configuration, inclusive of SIB19
satellite assistance information
note over e2nodes: Network initiated handover

@enduml
```

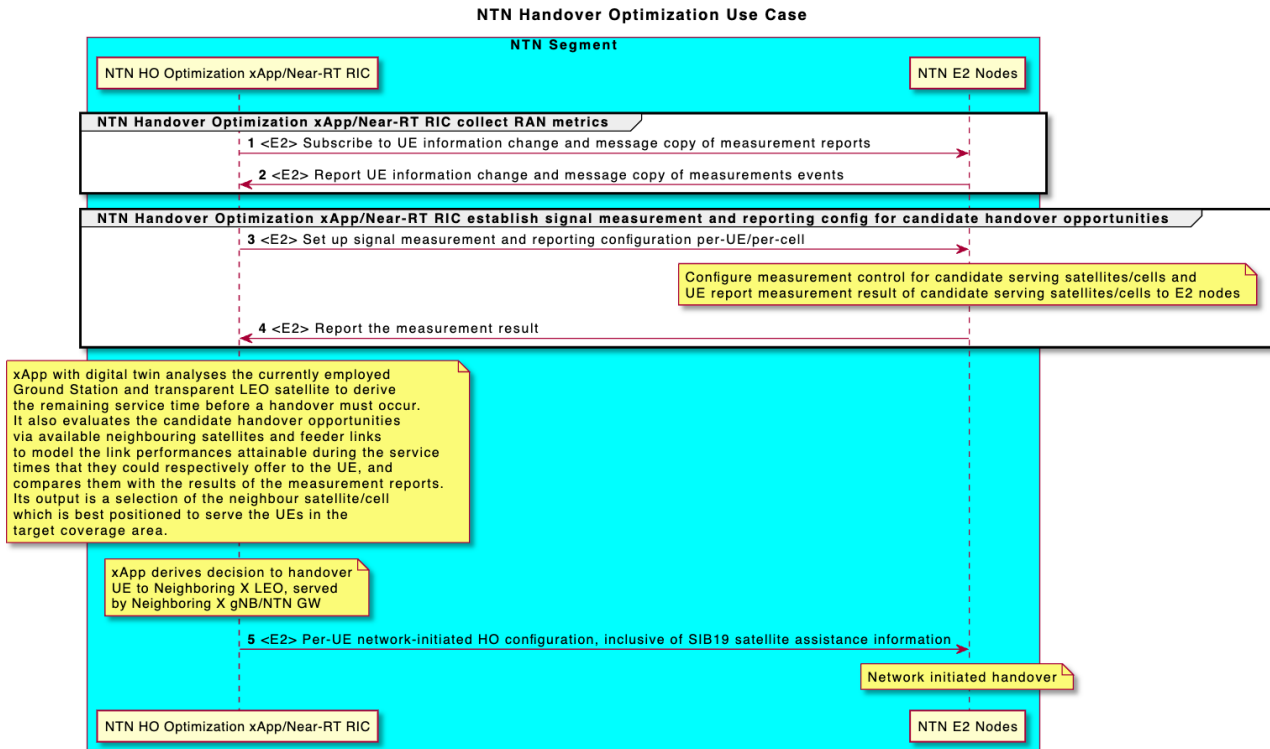



Figure 5.2.2.3-1: UML chart of NTN handover management

5.2.2.4 Benefits

The first and main benefit is the introduction of support of Release 17+ NR NTN network-controlled Handover (HO) from RRC Connected Mode, including support for NTN-specific satellite assistance information in the HO request/trigger, which is presently not supported by the O-RAN specifications.

The second benefit, achieved by the introduction of an NTN RIC (Near/Non-RT RIC) that is aware of the dynamic topology, physics and evolution of the constellation, and is capable of predicting its behaviour, is a potentially significant reduction of the signalling overhead, as well as a reduction of the potential for handover failures or sub-optimal handovers, leading to an improved service experience, as well as overall improved efficiency of the network by reduction of the resources consumed to support a mobility event.

5.2.2.5 Impact on O-RAN WGs

The use case brings an intelligent NTN handover management approach to O-RAN, which also utilizes information that is not natively exposed by the O-RAN interfaces (information owing to the non-3GPP components of an NTN constellation, and the constellation's own dynamic evolution).

There is a need to support the configuration of the previously described new measurement configurations specific to NR NTN introduced in Release 17, such as the new SSB Measurement Timing Configuration (SMTTC) or SMTTC4 (*SSB-MTC4List-r17*). In addition, the NTN cells must be able to broadcast Satellite Assistance information, in particular Satellite Ephemeris, in the form of SIB19, containing serving and neighbor satellite and cell information. Furthermore, in order to trigger a HO, the gNB must be able to include specific SIB19 content in the HO triggers.

These could be implemented by extending the existing support for measConfig and configuration of HO that is found in E2 Service Model RAN Control, as well as investigating new mechanisms either via E2 or O1 to deliver SIB19 information to gNB from the RIC/SMO.

5.2.3 Use Case 2-2: NTN QoS-based inter-cell traffic steering and load balancing

5.2.3.1 Use Case description

This use case covers the use of an NTN RIC, specifically a Near-RT RIC and xApp(s) to support the implementation of traffic steering and load balancing for administrative and network optimization reasons in an NTN network supporting NR NTN with satellite access via Release 17 and above. This use case is similar to the Traffic Steering Use Case described in the O-RAN Alliance WG1 Use Cases Analysis Report, however it focuses on Non-Terrestrial Networks (NTN) with satellite access and includes Load Balancing, as well as explicit awareness of orbits, orbital dynamics, cell load and QoS.

The current O-RAN use case definitions on Traffic Steering and QoS-aware scheduling do not take into account issues specific to NTN deployments, satellite or otherwise.

This use case's applicability is not limited to a specific orbit, and is instead applicable to any GSO or NGSO orbit, including but not limited to GEO, MEO and LEO. This use case can also apply to multi-orbit scenarios (e.g. GEO + LEO, GEO + MEO, etc), where different orbit types are specified in 3GPP as different RAT types. The scope of inter-RAT Traffic Steering and Load Balancing for this use case is presently focused on 5G NR RAT types where the RAT type is used to differentiate the NTN orbit.

5.2.3.2 Problem statement and challenges

It is well understood that in terrestrial cellular networks, imbalances in the traffic load across different cells, or differences in their available bandwidth and QoS capabilities, can lead to network hotspots, network congestion, suboptimal spectrum utilization and negative impact on the user experience due to for example increased latency.

It is also understood that networks may provide access to users with diverse profiles, as well as support different types of services and their associated requirements.

This challenge is tackled in the terrestrial domain by the RICs evaluating the QoS targets of the various DRBs/UEs as well as the current resource utilization in the network to determine whether there are sufficient resources allocated in the correct locations to meet the QoS requirements of users.

In NTN deployments, particularly those in Low Earth Orbit (LEO), the inherent movement of satellites means that components of the RAN, the air interface and even its transport might no longer traverse deterministic fibre optic connections, but rather wireless links between various space and ground assets of the NTN network which are likely to have time- and space-variant properties, such as high mobility, bandwidth constraints, high, variable or diverse delay, as well as potentially low reliability.

Even in 3GPP Release 17/18 specifications, which mostly focus on transparent/bent-pipe architectures, even though the entire RAN and the interconnections between its associated RAN Network Functions are placed on Earth, the NR Uu air interface has to traverse at least one transparent satellite hop via a radio feeder and user link, which can induce a series of time-variant impairments and latencies that impact the network's (and a RIC's and xApp's) ability to appropriately steer and load balance traffic in sync with the RAN scheduler.

It is also worth noting that traffic steering methodologies have been devised to achieve intelligent mapping of UEs to cells with an assumption that the cells might be at worst macro and micro cells, or more typically closely adjacent and similar tier cells. In the NTN context, on the other hand, the scenarios might involve networks with multiple orbits, where UEs in a given region could be split between GSO and NGSO segments, as well as within NGSO networks where different types of UEs in the same cell could be best served by different NGSO satellites depending on their specific connectivity requirements (e.g. eMBB vs URLLC) as well as the dynamic state of the network. This diversity of access layers may create significant differences in the latency, mobility, capacity, coverage area as well as several other factors.

Lastly, in terrestrial macro mobile networks, UEs with specialized characteristics and vertical-specific QoS requirements might be a comparatively small minority as opposed to eMBB smartphone users, at least in a given area of the network or band group. In NTN, on the other hand, this diversity is expected to be quite significant, both in terms of a mix of handheld (e.g. smartphone) and non-handheld UE (e.g. automotive, CPE, UAV, etc) even within the same cell, frequency band and RAT.

This makes it such that a number of aspects introduced by the nature of NTN systems, and that will be impacting the ability of the RIC to perform accurate traffic steering and load-balancing, are not yet accounted for in the O-RAN specifications, and thus require further analysis.

5.2.3.3 Proposed Solution

5G NTN aims to primarily serve UEs in rural outdoors environments where limited LoS obstruction on the ground, limited multipath effects and the predictable movement of the satellite in the sky coupled with accurate weather forecast data means that the channel quality can be more reliably modelled and estimated. Such a capability would allow an xApp/rApp solution to not only rely on instantaneous signal measurements, but also to perform modelling of the user and feeder link channel qualities as their link vectors change due to the orbital movement of the satellite. Additionally, by having access to information about the Satellite Transport Network such as the possible or currently employed antenna configurations onboard the satellites and the scheduled feeder link handovers, an xApp/rApp solution could obtain this cognisance of the downstream impacts of steering specific types of traffic and services in different ways.

For example, upon handover opportunities for a specific UE which are “forced” such as those that occur because the current serving satellite is dropping below the horizon, the xApp/rApp could perform channel modelling and predictions on the array of potential user and feeder link vector pairs which could be employed for that UE during the viable service time of that user-feeder link pair (i.e. prior to the next necessary user or feeder link handover). From this, it could derive which handover opportunity for a given UE/group of UEs would result in a service time during which the highest attainable spectral efficiency would be achieved on the user/feeder link pair, which would be the criterion on the basis of which DRBs and associated UEs with eMBB type of QoS flow targets would be managed. In parallel, for those UEs which are requesting a URLLC type of service, the xApp/rApp would also take into account any associated packet delay budget requirements which would have to be enforced, while also attempting to evaluate which of the candidate handover opportunities could deliver service the longest before either a user or feeder link handover would require temporary service disruption.

In cases of significant congestion, the xApp/rApp would prioritise those UEs and QoS targets which are regarded as being of highest priority, and would use other measures such as those defined in the existing QoS-based resource optimisation use case in WG3 UCR to find alternate solutions for those UEs whose QoS targets cannot be met in the current network state. To support the kinds of Conditional Handovers which are crucial for NTN mobility, as was highlighted in Use Case 2-3 in Clause 5.2.4, the current interface specification(s) would need to be extended. Lastly, if only a subset of potential satellites are relevant handover candidates for a UE consuming a specific type of service, the xApp/rApp solution would aim to decrease the overall power consumption and signaling overhead on behalf of the UE and RAN by constraining the SIB19 contents (e.g. ntn-NeighCellConfigList) and SMTC4 configuration to only produce Measurement Reports for those satellites which are determined to be relevant.

Subsequently, one of the handover capabilities defined in Use Case 2-3 in Clause 5.2.4 could be used to facilitate the handovers.

Table 5.2.3.3-1: NTN QoS-based inter-cell traffic steering and load balancing

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	Find best candidate serving satellites/cells to best meet QoS requirements for a variety of UEs while utilizing the least amount of radio resources in the network	
Actors and Roles	Non-RT RIC, NTN QoS-aware Traffic Steering and Load Balancing rApp, Near-RT RIC, NTN QoS-aware Traffic Steering and Load Balancing xApp, E2 Nodes.	
Assumptions	All relevant functions and components are instantiated. UE is already in the NTN segment and served by the NTN network. The NTN RIC is aware of the state and motion of the constellation and is capable of predicting its evolution in time and space.	
Pre-conditions	A1/E2 interface connectivity is established, QoS xApp has access to live source of global weather data	
Begins when	Operator specified trigger condition or event is satisfied.	
Step 1 (M)	Non-RT RIC sends the QoS optimization policy for UEs to Near-RT RIC via A1 interface.	
Step 2 (M)	NTN QoS Optimization xApp/Near-RT RIC subscribes to performance metrics, UE information change and message event of	

	UE mobility from E2 Nodes to understand the state of the network and location of UEs within it.	
Step 3 (M)	NTN E2 Nodes report performance metrics, UE information change and message event of UE mobility to NTN QoS Optimization xApp /Near-RT RIC.	
Step 4 (M)	NTN QoS Optimization xApp/Near-RT RIC performs analysis of collected information and determine the candidate serving satellites/cells with its digital twin or AI/ML model.	
Step 5 (O)	NTN QoS Optimization xApp/Near-RT RIC signals to NTN E2 Nodes to establish signal measurement and reporting in UEs for candidate handover opportunities	
Step 6 (O)	E2 nodes configures measurement control to UEs and UEs perform measurements and monitor for configured measurement events, upon which they signal a measurement report to the serving NTN E2 Node	
Step 7 (O)	NTN E2 Nodes forward the measurement report to NTN QoS Optimization xApp/Near-RT RIC	
Step 8 (M)	NTN QoS Optimization xApp/Near-RT RIC performs analysis of collected updated information and make the decision of handover to target serving satellites/cells with its digital twin or AI/ML model.	
Step 9 (M)	NTN QoS Optimization xApp/Near-RT RIC signals to NTN E2 Node to establish the correct handover control to the target UE	
Step 10 (M)	NTN E2 Nodes configure measurement control to UE for current serving satellite/cell.	
Step 11 (M)	NTN QoS Optimization xApp/Near-RT RIC continuously signals to Non-RT RIC feedback on enforcement status of defined QoS requirements	

```

@startuml qos optimization
title NTN QoS Optimization Use Case
Skin rose
skinparam sequenceMessageAlign left
skinparam ParticipantPadding 5
skinparam BoxPadding 10
skinparam defaultFontSize 12

box "SMO Framework" #gold
participant "Non-RT RIC" as nonrtric
end box

box "NTN Segment" #cyan
participant "NTN QoS Optimization xApp/Near-RT RIC" as xappnearrtric
participant "NTN E2 Nodes" as e2nodes
end box

autonumber

nonrtric -> xappnearrtric : <A1> Non-RT RIC sends the QoS optimization policy for UEs.

group #White NTN QoS Optimization xApp/Near-RT RIC collect UE information
xappnearrtric -> e2nodes: <E2> Subscribe to cell performance metrics, UE information change and message event of UE mobility

```

```

e2nodes -> xappnearrrtric: <E2> Report cell performance metrics, UE information change and message
event of UE mobility
end group
note over xappnearrrtric: Perform analysis according to current cell-level resource usage,\nper-
DRB QoS requirements and future-looking and weather-dependent\nlink budget analysis to derive
candidate per-UE handover opportunities

group #White NTN QoS Optimization xApp/Near-RT RIC establish signal measurement and reporting
for candidate handover opportunities
xappnearrrtric -> e2nodes: <E2> Set up signal measurement and reporting configuration per-UE/per-
cell
note over e2nodes: Configure measurement control for candidate serving satellites/cells and\nUE
report measurement result of candidate serving satellites/cells to E2 nodes
e2nodes -> xappnearrrtric: <E2> Report the measurement result
end group

note over xappnearrrtric : Take into account measurement reports to derive target handover
occasion
xappnearrrtric -> e2nodes: <E2> Per-UE network-initiated handover or CHO configuration
note over e2nodes: Network initiated handover
xappnearrrtric -> nonrrtric: <A1> QoS policy feedback

@enduml

```

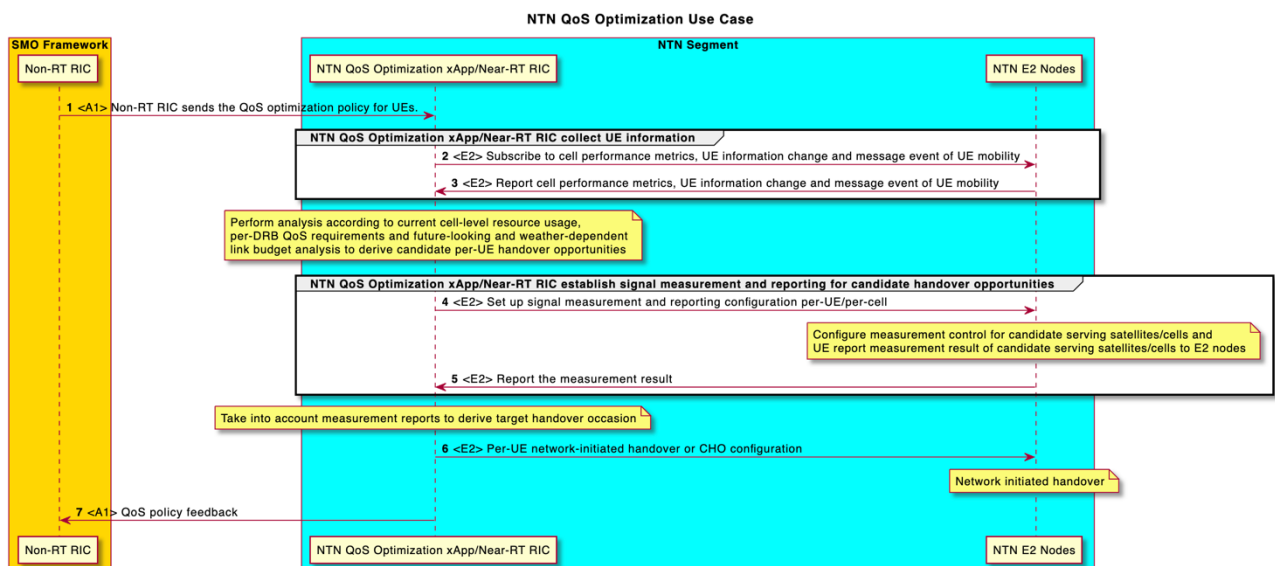


Figure 5.2.3.3-1 UML flow chart of use case of NTN QoS-based resource optimization and traffic steering

Sample required data:

1. QoS Optimization policy
 - a. Guaranteed Flow Bit Rate (GFBF)
 - b. Maximum Flow Bit Rate (MFBF)
 - c. Priority Level
 - d. Packet Delay Budget (PDB)
2. Metrics and information reporting
 - a. Performance metrics
 - i. Average DL/UL UE throughput
 - ii. Mean DL/UL PRB used for data traffic
 - iii. Packet Delay
 - iv. DL Synchronization Signal based Reference Signal Received Power (SS-RSRP)
 - v. DL Synchronization Signal based Signal to Noise and Interference Ratio (SS-SINR)
 - b. UE information changes
 - i. UE ID
 - ii. UE RRC state

5.2.3.4 Benefits

Employing limited network resources to communicate information is naturally at the centre of mobile networks, and especially critical for NTN. By leveraging access to network telemetry as well as employing accurate models of the behaviour of the system components and channel(s) to produce prediction-based data-driven decisions, the adoption of an O-RAN RIC and xApp-driven NTN control plane would lead to overall increase in efficiency of the system, lower spectral and power usage in the network, both from the aspect of achieving higher MCS on link vectors, as well as exhausting fewer resources.

This is expected to also lead to a higher overall rate of satisfied QoS and QoE requirements by increasing the number of fulfilled service targets with the same amount of resources, in turn improving the overall commercial viability of the SNO's NTN operations.

5.2.3.5 Impact on O-RAN WGs

The use case brings forth an NTN-specific angle and approach to Traffic Steering and QoS-based optimization use cases which were already addressed for terrestrial networks in WG2 and WG3. While the 5G QoS framework has not changed significantly for NTN, there is a need to support the configuration of NTN-specific network-controlled HO for traffic-steering and load-balancing purposes. For the SIB19 component, an opportunity exists for an xApp/rApp to update the relevant RAN nodes with some of the contents of the System Information Block 19 (e.g. Satellite Assistance Information and Ephemeris, neighbor satellite and cell configurations, k_{offset} , etc) which it ought to broadcast in the cell or unicast to specific users during HO command.

5.2.4 Use Case 2-3: NTN Time-based and Distance-based Conditional Handover (CHO)

5.2.4.1 Use Case Description

This use case covers the support of the new Time-based and Distance-based Connected Mode Conditional Handovers (CHO) introduced in 3GPP for NR NTN since Release 18.

The handover end-to-end call flow requires a sequence of signaling exchanges between the UE, its serving cell and a target neighboring cell to take place in order for the full procedure to be successful. In an event where the UE loses connectivity to its serving cell before it has received the required instructions to join the target cell, this could inhibit it from successfully undergoing a handover in RRC Connected Mode. Such an event is largely avoided in terrestrial networks by tailoring the signal measurement event on the serving cell to be triggered before its signal becomes too weak, but can become more likely to occur in 5G NTN constellations where the physical mobility of the satellites could mean that one which is currently delivering a cell to a specific location on the ground would all of a sudden become unable or ineligible to continue doing so, interrupting the procedure. This risk is only made higher by the increased control plane latencies witnessed in NTN networks, especially in transparent NTN systems where the RTT could be quite long, and when coupled with any additional processing time on the UE and RAN it might leave the UE(s) with insufficient time to receive the command to initiate the Random Access procedure prior to losing connectivity to the serving cell.

To account for such events, 3GPP has extended the capabilities of the Conditional Handover (CHO) with support for Time-based and distance-based CHO configurations in the UE, in which its serving cell can preauthorize the UE to initiate the handover to a neighbor cell under conditions which are precomputed, and which provide the UE(s) with sufficient opportunity to undergo a successful handover, while not attempting it either before the neighboring cell becomes available or unnecessarily early, the latter of which would increase the frequency of handovers which are a costly procedure. This functionality is particularly relevant in NTN because the difference in signal strength owed to the UE being located at the cell center versus the cell edge is not as pronounced as it would be in terrestrial networks, as discussed in Use Case 2-1 in Clause 5.2.2, and is an insufficient trigger to determine the necessity for a handover to take place. The Time-based and Distance-based triggers on the other hand do not account for what could be a potentially ephemeral drop in signal quality of the serving cell as more traditional signal measurement triggers might, but rather communicate that there is a hard limitation to the currently serving cell's ability to sustain service to the UE. By applying this configuration to all UEs in a cell, significant reductions in signaling overhead can be achieved compared to the regular handover scenario.

The Time-based and Distance-based CHO, introduced in 3GPP Release 17 as *condEventT1* and *condEventD1* in 3GPP TS 38.331 [1], have their applicability in both Earth-Moving Cell (EMC), and Earth Fixed Cell and quasi-Earth Fixed Cell (both addressed as EFC hereafter) applications, as discussed below.

5.2.4.3 Problem Statement and challenges

5.2.4.3.1 Time-based CHO

Accounting for Earth-Moving Cell based systems first, NTN constellations in NGSO employing the EMC configuration include satellites which project beams onto the Earth which are fixed to the body of the satellite itself, and are effectively sweeping over the Earth underneath. Even though the individual beams can be hundreds of kilometers in diameter, the high orbital velocity of a satellite that is for instance in LEO would mean that a UE would enter into the coverage of a single beam and exit it within the order to seconds (depending on factors like satellite elevation angle, angular beam width, satellite altitude, etc.).

This is exemplified in the Figure 5.2.4.3.1-6 below, showing a simplified representation of UE (orange point) mobility between cells illuminated by two NTN LEO satellites (blue and green points), applicable to both Transparent and Regenerative architectures, employing Earth-Moving Cells.



Figure 5.2.4.3.1-1: EMC inter-satellite handover scenario

In the first instance, the handovers occur between cells illuminated by the same satellite, exemplified as the blue satellite in Figure 5.2.4.3.1-1. In a Regenerative architecture, these would most often involve cells which belong to the same onboard gNB and would constitute intra-gNB handovers. In Transparent architectures, the same could be the case (successive cells belong to same gNB on the ground), or the satellite could be connected via multiple feeder links to different gateways and gNBs, in which case successive beams could carry cells delivered by different gNBs and would therefore constitute an inter-gNB handover. Nevertheless, eventually the UE would be covered by one of the “outer” beams of a single satellite after which there is no further beam arriving from the same satellite. In this event, there needs to be another satellite coming either in succession in the same orbital plane or from another crossing orbital plane (the green satellite in the same Figure) which can resume service to the UE with a new beam and cell. Again, depending on the architecture employed, this could either lead to an intra- or inter-gNB handover. As discussed before, signal measurement-based handover mechanisms would be insufficient to drive this type of mobility. Because of the frequency of handovers and the potential high number of UEs on the ground, significant signaling overheads would be generated as the UEs are being swept by different beams and cells. Time-based CHO was introduced in 3GPP to address exactly this.

Even if the satellites in the NTN constellation are capable of beam steering, allowing them to illuminate and track Earth-Fixed Cells, there is still a fundamental limitation whereby a satellite in LEO would eventually fall beneath the minimum elevation angle, causing it to stop illuminating a 5G cell over the EFC, and the EFC would need to be handed over to a 5G cell served via a subsequent satellite. The two satellites could provide overlapping service with their respective cells for a short period of time during the handover of the EFC from one to the other, but they need not to, and the coverage can even be discontinuous in time as long as the associated timers (e.g. T304, T310) do not expire before the UEs in the EFC connect to the subsequent cell, causing RLF. This type of handover of entire EFCs (and constituent UEs) is applicable to Transparent architectures where different gNBs on the ground deliver successive cells via different gateways and satellites, where the same gNB delivers successive cells via different gateways and satellites, as well as in cases where two different gNBs via different gateways form feeder links with the same Transparent satellite that illuminates both of the successive cells onto the same EFC. It is also applicable to Regenerative architectures where the gNB is placed onboard the satellite, and the two gNBs/satellites deliver successive cells to the same EFC.

The Figure 5.2.4.3.1-2 below showcases a simplified view of the above scenarios from the point of view of the satellite user links (omitting gateways and feeder links), whereby the orange UE is initially served by a cell delivered via the blue satellite (PCI 3). As the blue satellite is progressing along its orbit due North-East, it is nearing the moment where it will no longer be capable of illuminating and tracking the EFC in which the orange UE is located. At the same time, the red LEO satellite is

coming into view, moving due South-East, and can resume the service to the EFC. It would deploy a new cell on top of the EFC (PCI 13), either with overlapped or discontinuous service.



Figure 5.2.4.3.1-2: EFC inter-satellite handover scenario

Yet again, the need for a handover is not driven by variable signal quality by serving and candidate neighboring cells, but either by a hard limitation to the serving cell's ability to sustain service to the EFC in question, or a higher layer decision by an NTN optimization xApp to redirect the blue satellite's beam and power resources to provide coverage onto another EFC, even if the blue satellite could sustain coverage to the original EFC for a longer period of time. While UEs in EFCs would nominally undergo handovers less frequently compared to UEs in EMCs, risks owing to the reliability of just-in-time control plane signaling delivery and overall impacts on signaling overhead can be mitigated through the employment of Time-based CHO to drive handovers, where the exact moment of handover can be precomputed and delivered ahead of time to the NTN gNB, who would signal it to the UEs in the EFC in question.

5.2.4.3.2 Distance-based CHO

While the ephemeral nature of satellite coverage over a given region defines the time bounds between which UEs in the region in question can be served, there are also other determinants which might cause a UE to exit coverage of one cell or beam, and necessitate a handover to another cell or beam, if present. One of these determinants is spatial in nature; namely, a UE's physical location with respect to the spatial coverage footprint of a satellite beam via which the serving cell is illuminated will determine if the UE can continue to be served by that cell. By the same logic, a UE's physical location with respect to that of a neighboring beam and cell will determine if the UE can be handed over to that cell.

Focusing on EMC-based systems first, this determinant can be exemplified by reusing Figure 5.2.4.3.1-1. With considerations on whether the architecture is Transparent or Regenerative being equal to the Time-based CHO EMC case, the coverage beams of NGSO satellites for which the EMC scenario applies are often approximated using hexagons, but their illumination footprints are in practice more circular in nature, with overlapping coverage towards the circular beams' edges. Therefore, the footprint area can be nominally defined using the circular footprint's center and radius (this holds true best when satellites are directly at zenith over a coverage region; at lower elevation angles the beam footprint illuminates the Earth's surface at an angle, causing the effective footprint to be more oval). Focusing on Figure 5.2.4.3.1-1, the orange UE is initially inside the satellite beam and cell with PCI 3. As the blue satellite traverses overhead along its orbital path, at one point the orange UE, even if stationary, will find itself more than the radius distance away from the moving center of the cell with PCI 3 (implicitly exiting its coverage), and less than the radius distance away from the moving center of the neighboring cell with PCI 4 (implicitly entering its coverage). With the expected slight overlap of the beams, there is a short period of time during which the UE can be expected to be in coverage of both cells. As was the case with Time-based CHO, relying on signal measurements from the serving and neighboring cell(s) to drive such mobility cases is insufficient, and the legacy handover procedure can be expected to meaningfully contribute to signalling overhead and reduce the robustness of the handover procedure in an NTN scenario. For this purpose, the Distance-based CHO mechanism was extended in 3GPP Release 18 to support EMC cases.

The Distance-based CHO mechanism was in fact originally introduced in 3GPP Release 17 in support of EFC-based systems. In contrast to EMC, satellite beams and cells are fixed to the Earth and so, while their footprints can still be approximated as circles with a center and radius, the footprints' centers are stationary. Nevertheless, the UEs themselves could be moving, and with them their distance from the serving and neighboring cell centers could be changing. At one point, a UE will have moved outside the effective coverage of its currently serving cell, and into that of another neighboring cell, if present, leading to a handover event.

Taking Figure 5.2.4.3.2-3 as an example scenario, the orange UE is initially located inside the blue cell with PCI 6. After some time, due to the UE's mobility, it will have travelled due North and crossed over into the coverage of the blue cell with PCI 3. At this point, the UE becomes closer to the center of the cell with PCI 3 than to that of the cell with PCI 6. Similar to the explanation for EFC in Time-based CHO scenarios, the blue satellite could either be Regenerative and the handover between two blue cells would be an intra-gNB handover, or the blue satellite could be Transparent, potentially forming one or more feeder links to one or more gNBs, depending on which the handover could be intra- or inter-gNB.

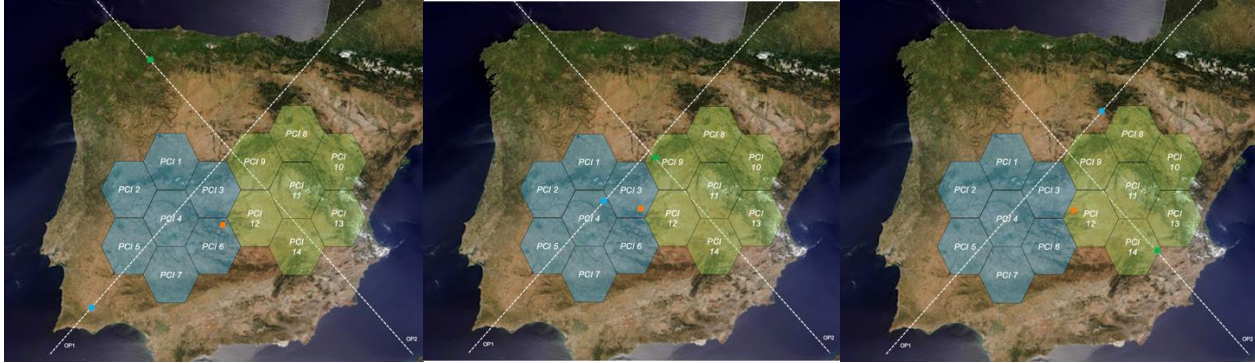


Figure 5.2.4.3.2-3: EFC inter-beam handover scenario

Once in the blue cell with PCI 6, after some time the orange UE might move Eastwards towards the area which is covered by the green cell with PCI 12, coming from the green satellite. Once more, the overall architecture at play would influence what type of handover this is. If the blue and green satellites are Regenerative, this would be an inter-gNB handover while if they are Transparent, the same could be the case or both might even be connected via feeder links to the same gNB, implying an intra-gNB handover.

The current E2SM-RC [2] specification does not presently support Time-based and Distance-based measurement and Conditional Handover events (e.g. *condEventT1*, *condEventD1* and *condEventD2* in TS 38.331[1]) required for the Near-RT RIC to implement the new NTN-specific Time-based and Distance-based Conditional Handover (CHO) procedures introduced by 3GPP in Release 17 and Release 18.

As described previously, this can lead to a high signaling load, especially in Earth-Moving Cells (EMC) systems, whenever the satellite exits the coverage area and all the UEs within the cell need to be handed over. For high-speed UEs (e.g. aircraft, fast UAV), due to the signal profile in an NTN cell, this could lead to the call being dropped in case the traditional HO procedure cannot be sufficiently responsive as the UE reaches the cell edge, causing it to hit Radio Link Failure (RLF).

5.2.4.4 Proposed Solution

5.2.4.4.1 Time-based CHO

Addressing the EMC case first, whether the system is Transparent or Regenerative, 3GPP has not provisioned or accounted for the NTN gNBs to have the ability to keep track of which satellites, and by proxy which EMC cells, are valid neighboring candidates for handover in a continuously evolving system such as an NGSO NTN constellation. This is exemplified in the simplified Figure 5.2.4.3.1-6 but would be even more pronounced in a denser deployment, as it would be in a real constellation.

The blue satellite in the Figure, orbiting due North-East along its orbit, is the one which is initially serving the orange UE via the cell with PCI 3. Following 3GPP Release 17, the NTN gNB, as prompted by the NTN xApp via the E2 interface, can instruct the UE with GNSS capabilities to record coarse location information (with ~2 km accuracy) and report it to the NTN gNB [1], who subsequently uses the E2 interface to deliver it to the NTN Near-RT RIC and the NTN xApp. By knowing the coarse coordinates of the UE as well as using the digital twin to derive the coverage footprints of the individual cells/beams delivered by the blue satellite, the NTN xApp can compute the remaining service time of the cell with PCI 3 after which the UE needs to be handed over to the cell with PCI 4, and subsequently to the cell with PCI 5. These derived remaining service times and handover target cells are UE-specific even within the same cell (e.g. PCI 3) as the relative location of different UEs within the same cell would define their mobility opportunities. For instance, if the UE in Figure 5.2.4.3.1-6 was instead in the North-West corner of the cell with PCI 3 in the initial view, its target handover cell would instead be the cell with PCI 1, and the handover would need to occur sooner due to the shape of the beam and cell with PCI 3.

The NTN xApp would instruct the NTN gNB delivering the cell with PCI 3 to configure the orange UE with the T1 CHO (*condEventT1* [1]) conditions that would signal to it the time interval during which it ought to attempt to connect to the cell that is subsequent to it, the one with PCI 4, as well as with appropriate SIB19 parameters to be broadcast that would contain the *ntn-Config-r17* of all of the neighboring cells, itself containing the *epochTime-r17* and *ephemerisInfo-r17* of the blue satellite, among other parameters. Taking the cell with PCI 3 from the above Figure as an example, its neighboring cells would be cells with PCI 1, PCI 4 and PCI 6.

After some time, the orange UE enters the trailing beam/cell of the blue satellite (PCI 5) after which it needs to be handed over to a cell illuminated by another satellite, in this instance the green one (first cell being one with PCI 14). The NTN Near-RT RIC and NTN xApp would yet again combine coarse UE location information and digital twin modelling of the NTN constellation to produce the T1 CHO (*condEventT1*) and SIB19 parameters which aid the handover from blue cell with PCI 5 to the green cell with PCI 14. This could be coupled with another condition, such as one that is dependent on the signal quality of the neighboring cell exceeding some nominal value to ensure that the handover stands a greater chance of success (e.g. *condEventA3*, *condEventA4*, *condEventA5*) for which the SMTC4 configuration defined for NTN is employed (or otherwise can be used to pre-synchronize with the neighboring SS/PBCH transmissions). Furthermore, the overlapping coverage from multiple satellites also necessitates close coordination of which frequencies are employed by which cell in order to avoid interference, and the question of which are permitted to be employed is in many cases dependent on the region of the Earth that is being illuminated. Therefore, there is a further need for a higher order NTN control function, such as an NTN Near-RT RIC and constituent NTN xApp(s), to drive this functionality.

As for the EFC case, in good time prior to the scheduled discontinuation of coverage to the EFC by the blue cell with PCI 3, the NTN Near-RT RIC and NTN xApp would instruct via the E2 interface the NTN gNB to configure the UEs in the cell with the appropriate T1 CHO (*condEventT1*) configuration, potentially in conjunction with signal measurement-based CHO configurations (e.g. *condEventA3*, *condEventA4*, *condEventA5*). As was the case in the EMC instance, the NTN xApp will infer, if not decide itself, through the NTN digital twin and forward-looking scheduling that it will be the red satellite which will deploy a particular cell (PCI 13) at a particular moment over that EFC in order to continue providing service, and will have therefore provisioned the T1 CHO configurations in good time prior to the handover occasion taking place, taking into account any dependency on signaling delays induced by the RTT influenced by the orbital regime and Transparent vs. Regenerative nature of the constellation. Likewise, this will need to be accompanied with the updated SIB19 containing the *ntn-Config-r17* which will indicate to the UE crucial information about the cell with PCI 13 and the red satellite, such as its ephemeris information, enabling the UE to synchronize with it.

In the event that there is a significant number of UEs in the blue cell with PCI 3, their handing over to the red cell with PCI 13 during the potential overlap period should be managed carefully in order to reduce the chances of preamble collisions, contention resolution delays, etc. Since the two satellite beams and cells in question are intended to cover the same EFC, the relative location of different UEs within the EFC would not play as much of a role as it did in the EMC case, as all of the UEs would be affected by the withdrawal of the blue cell with PCI 3 and illumination of the red cell with PCI 13 in the same manner and time (there is an edge case where this need not be true; if the elevation angles of the blue and red satellites cause their beam footprints and therefore cell coverages on top of the EFC at the moment of handover to be of different shapes, excluding certain UEs). In this respect, the same T1 CHO parameters could be configured for all of the UEs within the same cell. Nevertheless, in order to balance the RA load onto the red cell with PCI 13, the NTN Near-RT RIC and NTN xApp(s) can theoretically alter the duration of any overlap or discontinuity of service to the EFC (T310 timer for RLF in UEs starts as blue cell shuts off – UEs must perform RA to the red cell prior to expiry to avoid exiting RRC Connected Mode) as well as tune the per-UE group RACH-Config and T1 CHO parameters to better distribute their RA across the red cell's RACH occasions.

5.2.4.4.1 Distance-based CHO

Focusing again on the EMC case first, and with respect to the scenario in Figure 5.2.4.3.1-6, the UE is initially served by the blue satellite via the beam and cell with PCI 3. It requires a mechanism by which to evaluate its distance from the moving locations which represent its current serving cell and a candidate neighboring cell in an EMC system. In 3GPP Release 18, a new CHO configuration was introduced, called D2 CHO (*condEventD2* [1]), for this purpose. This allows the NTN xApp, by proxy of the E2 interface to the NTN gNB, to specify two distances, *distanceThreshFromReference1-r18* and *distanceThreshFromReference2-r18* alongside a hysteresis distance, to a UE in a cell. The UE is expected to use its GNSS capabilities to ascertain its location, but it also needs the two moving locations to measure and compare its distance to.

At the same time, the NTN xApp can use the E2 interface to deliver to the NTN gNB the parameters which it is supposed to broadcast to UEs in cell with PCI 3 as part of SIB19. Other than including crucial parameters such as *epochTime-r17* and *ephemerisInfo-r17* which are found in the *ntn-Config-r17*, starting from 3GPP Release 18 the SIB19 is also supposed to also include a field called *movingReferenceLocation-r18*. This is supposed to signal to the UEs a serving cell-specific reference location which they are supposed to measure their distance to, and compare it to the *distanceThreshFromReference1-r17* value

provided via the D2 CHO configuration. Nominally this location can be regarded as representing the moving cell center, but strictly speaking need not be, and it is up to the NTN xApp to define what this is going to represent. The *movingReferenceLocation-r18* is in its definition a static point on the ground, timestamped to the *epochTime-r17* from the *ntn-Config-r17*, and the UE is expected to combine this with the *ephemerisInfo-r17* to propagate the satellite's orbital trajectory in the immediate future and ascertain how the *movingReferenceLocation-r18* would move along with the satellite, before it receives the subsequent SIB19 broadcast with the network-updated *movingReferenceLocation-r18*.

The other reference location, the distance to which the UE is measuring using its GNSS-derived coordinates and comparing to the *distanceThreshFromReference2-r17* value provided via the D2 CHO configuration, is delivered inside the *measObjectNR*, namely in the *ntn-NeighbourCellInfo-r18* field, for a specific candidate neighboring cell. Within this field, the neighboring cell's *epochTime-r17*, *ephemerisInfo-r17*, and *referenceLocation-r17* are specified, where the latter is a static location which indicates the reference location at *epochTime-r17*, which the UE can use alongside the *ephemerisInfo-r17* to propagate the satellite's orbital trajectory in the immediate future and ascertain how the *referenceLocation-r17* would move along with the satellite, as was the case with *movingReferenceLocation-r18*.

Once the distance to the *movingReferenceLocation-r18* of the serving cell exceeds the *distanceThreshFromReference1-r17*, and simultaneously the distance to the *referenceLocation-r17* of a neighboring candidate cell drops below the *distanceThreshFromReference2-r17* (taking into account hysteresis in both cases), the UE regards itself as having satisfied the conditions for the D2 CHO, and attempts to join the neighboring cell. Nominally, the NTN xApp does not need to know the coarse location of the UEs in a cell to supply them with the D2 CHO parameters, making the D2 CHO configuration common to all UEs in the cell. Referring to the scenario in Figure 5.2.4.3.1-6, if the UE were to be initially located inside the blue cell with PCI 4, it would be supplied with D2 CHO configurations only for the neighboring blue cells which could at some point in the future (no farther than the *TimeToTrigger* in the D2 CHO config) illuminate the current footprint of the cell with PCI 4. These would be the cells with PCIs 2, 5 and 7, but not the ones with PCIs 1, 3 and 6, since those wouldn't be valid handover candidates due to the movement of the blue satellite. Then, if the serving and various neighboring cell reference locations were tied to the cell centers, the distances supplied in the D2 CHO config would relate to the serving and neighboring cells' radii.

Moving on to EFC-based systems next, and taking the scenario from Figure 5.2.4.3.2-2 as reference, the orange UE is initially located within the blue cell with PCI 6. The NTN xApp is aware of the overall EFC topology of the network, and can infer the cells which are neighbors to the cell with PCI 6. In the simplified example in the Figure, these are the blue cells with PCIs 3, 4, 7 and the green cell with PCI 12. In this instance, the cell centers are not influenced by the movement of the blue and green satellites, but are instead fixed on the ground. For supporting mobility cases where the UE's relative proximity to these reference locations is driven by the mobility of the UE itself, 3GPP introduced in Release 17 support for D1 CHO (*condEventD1* [1]). Within it, the two reference locations (*referenceLocation1-r17* and *referenceLocation2-r17*) and the distance thresholds to the two reference locations (*distanceThreshFromReference1-r17* and *distanceThreshFromReference2-r17*) are defined right in the D1 CHO configuration, and are common for all UEs in the cell. Referring again to the starting position of the orange UE, it receives a D1 CHO configuration for each of the neighboring cells. After some time, the UE moves due North sufficiently such that its distance from the center of the serving cell with PCI 6 (*referenceLocation1-r17*) exceeds the configured *distanceThreshFromReference1-r17*, while its proximity to the center of the neighboring cell with PCI 3 (*referenceLocation2-r17*) drops below the configured *distanceThreshFromReference2-r17*, taking into account hysteresis for both distances. At that point, the D1 CHO configuration for cell with PCI 3 is triggered, and the orange UE attempts to join it, where upon success it becomes configured with the D1 CHO configurations for the new set of neighboring cells (blue cells with PCIs 1,4 and 6 and green cells with PCIs 9 and 12). The triggering condition could again be coupled with another condition, such as one that is dependent on the signal quality of the neighboring cell exceeding some nominal value to ensure that the handover stands a greater chance of success (e.g. *condEventA3*, *condEventA4*, *condEventA5*) for which the SMTC4 configuration defined for NTN is employed (or otherwise can be used to pre-synchronize with the neighboring SS/PBCH transmissions). Yet again, after some time, the orange UE moves, this time Eastwards into the coverage footprint of the green cell with PCI 12. Here, the same conditional measurement procedure takes place, and the orange UE eventually joins the cell with PCI 12.

5.2.4.5 Benefits

This would enable the new NTN-specific Time-based and Distance-based Conditional Handover (CHO) procedures introduced by 3GPP in Release 17 and further refined in Release 18. These types of handovers were created specifically for their potential for reduction of the signaling overhead in a cell, by adding flexibility to when the network can trigger UE handovers, allowing the control signaling to be spread over time or sent during low-traffic intervals, in turn improving the user experience and smoothness of mobility. This is especially important for NTN systems with Earth-Moving Cells (EMC) and for fast-moving UEs.

5.2.4.6 Impact on O-RAN WGs

The support for the above solutions would require the introduction of support in the O-RAN E2 interface for the new NTN-specific Time-based and Distance-based Conditional Handover (CHO) procedures introduced by 3GPP in Release 17 and further refined in Release 18 (*condEventT1*, *condEventD1*, and *condEventD2*), with their corresponding new measurement events (*eventD1* and *eventD2*), as defined in TS 38.331 [1]. These would be further augmented by the NTN xApp being able to use the E2 interface to further:

- specify SIB19 parameters to be broadcast by a cell,
- specify SMTC4 (*SSB-MTC4List-r17*) parameters,
- specify the *ntn-NeighbourCellInfo-r18* information sent via the *measObjectNR*,
- configure recording of coarse-grained information about the UEs' GNSS-derived location,
- subscribe to reports of coarse-grained information about the UEs' GNSS-derived location

The above changes are anticipated to be best hosted in the O-RAN E2SM-RC [2] specification. Further to changes to the E2 interface, an Near-RT RIC optimized for hosting xApps purpose built for NTN optimizations is expected to be required to also expose to the xApps access to an NTN digital twin which the xApps can query to obtain the dynamic and evolving nature of the NTN constellation, including non-3GPP assets such as ground stations and the satellite buses themselves. In addition, such a Near-RT RIC could be required also to expose to the xApps an updated dataset of nowcast and forecast global weather data, which the xApps could be used to calculate prospective radio link conditions if different configurations of the NTN constellation were employed to deliver connectivity to the ground.

5.2.5 Use Case 2-4: NTN Idle mode cell re-selection

5.2.5.1 Use Case Description

Void.

5.2.5.3 Problem Statement and challenges

Void.

5.2.5.4 Proposed Solution

Void.

5.2.5.5 Benefits

Void.

5.2.5.6 Impact on O-RAN WGs

Void.

5.2.6 Use Case 2-5: NTN UE Location-aware Connected Mode Handover and Inter-cell Traffic Steering and Load-balancing

5.2.6.1 Use Case description

This use case describes leveraging of knowledge of UE location (coarse or fine) by the NTN network to aid mobility (e.g. HOs), as well as other network control procedures such as inter-cell Traffic Steering and Load Balancing.

5.2.6.2 Problem statement and challenges

Normally, in a mobile network it is assumed that handover decisions are taken without awareness of the UE's location in the network. This is typically sufficient in terrestrial networks because the cells, even macro cells, are in general reasonably small and, in MIMO and other beamforming systems, the spatial granularity of the UE positions within the cell is within the resolution of a single gNB beam in the beamforming matrix. As the number of UE per cell densifies, a typical solution is to aim for ever larger MIMO arrays, thus increasing the spatial resolution (e.g. Massive MIMO).

In addition, heavy multipath environments typical of terrestrial urban networks help the MIMO system by creating very distinct channel transfer functions between each user and the network.

This however is mostly not true for NTN, where each NTN node (e.g. a satellite) should be seen as a single large multi-beam MIMO array, and where each NTN beam is expected to cover a very large geographic area (e.g. several km, 10s or even 100s of km) and thus several UEs or groups of UEs. This has also the collateral effect of creating very large overlap areas between beams/cells, which, coupled with a typically "flat" cell signal distribution per-beam, makes it such that the cell boundaries are more complicated to establish. In addition, the NTN node motion makes it such that, as the NTN node, such as a satellite, flies over the service area, the signal level and quality will naturally vary quite significantly even as seen by a stationary UE.

This creates a number of gaps and problems, such as:

1. Inability of the network to precisely assess when a UE is approaching edge of cell (e.g. due to satellite mobility in EMC scenarios, or UE mobility in all other cases), which can make it difficult to assess whether a UE is about to require handover to an adjacent beam/cell.
2. Ambiguity in assessing the UE mobility and the signal reporting state across multiple beams/cells, as a UE may report a signal drop even when stationary, due to satellite movement, which can lead to ping-pong effects by making the network wrongly think that a UE is approaching the threshold at which it can be better served by an adjacent cell, beam or NTN node.
3. Inability of the network of discerning in which portions of the NTN beam (and thus potentially cell) UE are located or clustered, which may force a worst-case resource assignment, and may make it impossible (or very difficult) for the network to assess whether some UE could be offloaded to an adjacent cell/beam or NTN mode (e.g. satellite) to smooth cell load.

All of the above knowledge gaps can create problems and inefficiencies with both mobility, as well as resource assignment and traffic load-balancing at the network level. In all such cases, knowledge of UE locations or clustering, at least at coarse-grained level, by the NTN network, can significantly improve mobility reliability and resource management decisions.

Whilst some of the procedures are technically available since at least Release 16, new NTN-specific procedures have been introduced in Rel-18, and moreover there may be gaps in the O-RAN specifications to prevent full leveraging of existing solutions. Therefore the exact O-RAN specification gaps need to be further investigated.

5.2.6.3 Proposed Solution

Several mechanisms are available for the network to become aware of UE locations, at either coarse- or fine-grained degree.

For example, once the UE is in RRC_CONNECTED mode and AS security has been established, the network can request the UE to report its GNSS-derived location information, either coarse or fine-grained, through location management procedures either directly to the gNB, or via the Core Network (e.g. LMF).

Table 5.2.6.3-1: NTN UE-location-aware handover management

Use Case Stage	Detailed Information
Goal	Find best handover candidate satellites/cells for a UE while utilizing the least amount of radio resources in the network
Actors and Roles	NTN xApp, Near-RT RIC, E2 Nodes, UE.
Assumptions	All relevant functions and components are instantiated.
Pre-conditions	E2 interface connectivity is established, NTN xApp has access to live source of global weather data

Step 1 (M)	NTN Handover Optimization xApp/Near-RT RIC subscribes to UE GNSS, UE information change and message copy of UE mobility from E2 Nodes to understand state of the network and location of UEs within it.
Step 2 (M)	NTN E2 Nodes report UE GNSS, UE information change and message copy of UE mobility to NTN QoS Optimization xApp /Near-RT RIC.
Step 3 (M)	E2 Nodes configure measurement control to UEs and UEs perform measurements and monitor for configured measurement events, upon which they signal a measurement report to the serving NTN E2 Node
Step 4 (M)	NTN E2 Nodes forward the measurement report to NTN Handover Optimization xApp/Near-RT RIC
Step 5 (M)	xApp with digital twin analyses the currently employed Ground Station and transparent LEO satellite to derive the remaining service time before a handover must occur. It also evaluates the candidate handover opportunities via available neighbouring satellites and the feeder links they form with Ground Stations that host NTN gNBs to model the link budgets attainable during the service times that they could respectively offer to the UE, and in doing so it selects the neighbouring satellite/cell which is best positioned.
Step 6 (M)	NTN Handover Optimization xApp/Near-RT RIC signals to NTN E2 Node to establish the correct handover control to the target UE
Step 7 (M)	NTN E2 Node implements RRC Connected Mode handover as specified by NTN Handover Optimization xApp/Near-RT RIC

```

@startuml handover optimization
title NTN Handover Optimization Use Case
Skin rose
skinparam sequenceMessageAlign left
skinparam ParticipantPadding 5
skinparam BoxPadding 10
skinparam defaultFontSize 12

box "NTN Segment" #cyan
participant "NTN HO Optimization xApp/Near-RT RIC" as xappnearrrtric
participant "NTN E2 Nodes" as e2nodes
end box

autonumber

group #White NTN Handover Optimization xApp/Near-RT RIC collect RAN metrics
xappnearrrtric -> e2nodes: <E2> Subscribe to UE GNSS, UE information change and message copy of measurement reports
e2nodes -> xappnearrrtric: <E2> Report UE GNSS, UE information change and message copy of measurements events
end group

group #White NTN Handover Optimization xApp/Near-RT RIC establish signal measurement and reporting config for candidate handover opportunities
xappnearrrtric -> e2nodes: <E2> Set up signal measurement and reporting configuration per-UE/per-cell
note over e2nodes: Configure measurement control for candidate serving satellites/cells and\nUE report measurement result of candidate serving satellites/cells to E2 nodes
e2nodes -> xappnearrrtric: <E2> Report the measurement result
end group

note over xappnearrrtric
xApp with digital twin analyses the currently employed
Ground Station and transparent LEO satellite to derive
the remaining service time before a handover must occur.

```


It also evaluates the candidate handover opportunities via available neighbouring satellites and the feeder links they form with Ground Stations that host NTN gNBs to model the link budgets attainable during the service times that they could respectively offer to the UE, and in doing so it selects the neighbouring satellite/cell which is best positioned.

end note

note over xappnearrrtric : xApp derives decision to handover\nUE to Neighboring X LEO, served\nby Neighboring X gNB/NTN GW

xappnearrrtric -> e2nodes: <E2> Per-UE network-initiated handover or CHO configuration
note over e2nodes: Network initiated handover

@endum1

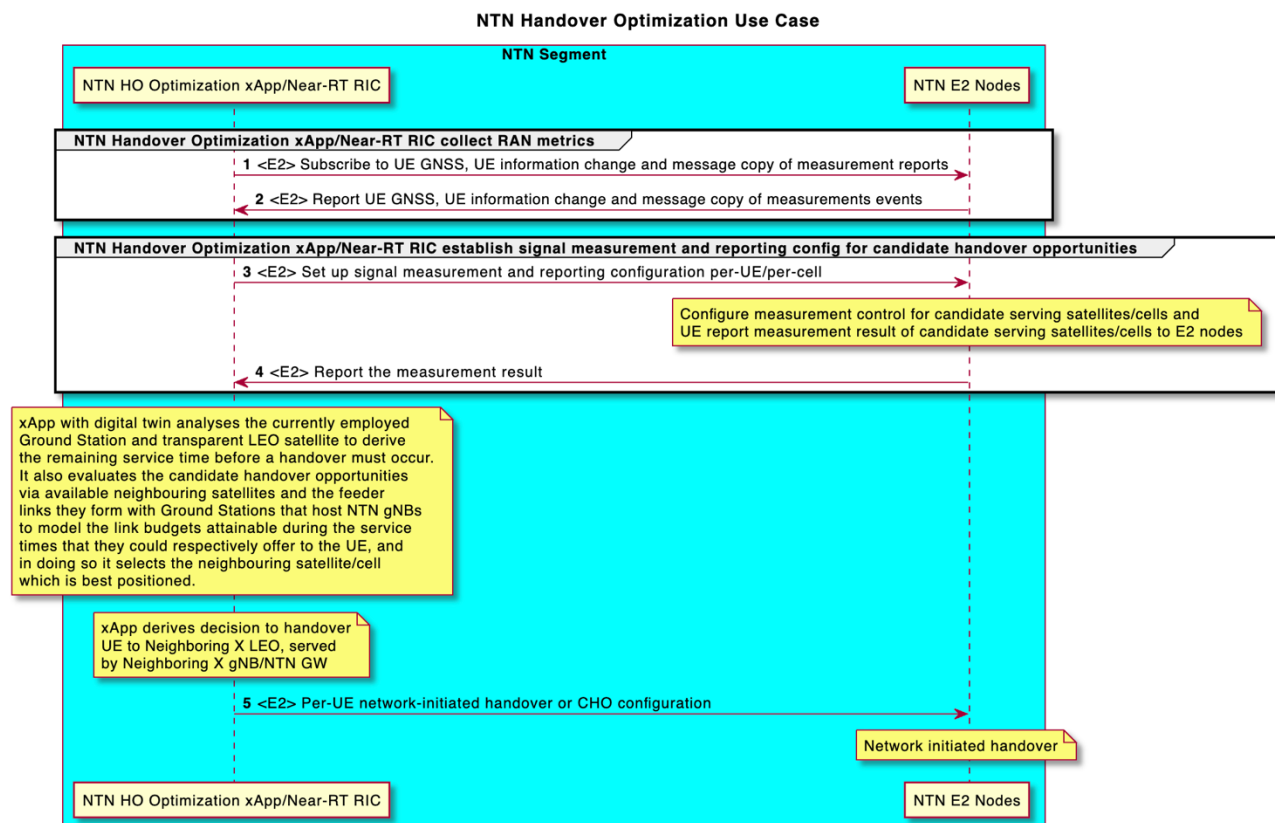


Figure 5.2.6.3-1 UML flow chart of use case of NTN UE Location-aware Connected Mode Handover and Inter-cell Traffic Steering and Load-balancing

5.2.6.4 Benefits

Additionally, the ability for the network to retrieve coarse GNSS location from the UE can help the NTN xApp augment its ability to assess the candidate link vectors for a particular UE.

5.2.6.5 Impact on O-RAN WGs

The current O-RAN E2 specification already supports retrieval of UE location, however, for NTN, there is also a need to enable retrieval of the new parameter for Coarse UE Location introduced in Rel-17 and 18.

6 Summary and Conclusion

Void

Annex:

Change history/Change request (history)

Date	Revision	Description
2024.12.12	00.00.03	Initial TR skeleton
2025.02.24	00.00.04	- Add introduction and NTN development status in standard organizations - The following CRs have been implemented: - KEY, AAL - CR 0001 – Introduction - v2 - KEY, AAL, FRN – CR 0002 - NTN development status in standard organizations - v3
2025.06.17	00.00.05	- Introduction of initial Use Case Group 2 set - The following CRs have been implemented: - AAL, KEY, RAD – CR 0004 - Introduction of Use Case Group 2 - v0.6_NOK_AAL
2025.06.18	00.00.06	- Update clause 1-3 and change history - introduction of Use Case Group 1, the following CRs have been implemented: - KEY, AAL – CR 0005 - fix clause 1-3 and change history - v3 - AAL, KEY, RMI, MTR – CR 0003 - Introduction of Use Case Group 1- v0.12_AtMeeting
2025.06.30	00.00.08	- Updates to the overall TR - Corrections to image references after first round of comments.
2025.07.09	00.00.09	Editorial and formatting for first release
2025.07.21	00.00.10	- Implementation of voting round editorial comments - Addition of CR references to the Change history/Change request (history)
2025.07.22	01.00	First formal release
2025.10.17	01.00.01	- Merge CRs for update of use case group 2 as below: - AAL, KEY, RAD CR-0005 - AAL, KEY CR-0007 - Fixed figure and table index
2025.12.02	01.00.02	Fix figure and table index and address comments in review
2025.12.02	02.00	November train 2025 release